

Lecture Notes

Professional Practices

Starting Out With Programming
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Lecture 1: Introduction to Professional Ethics, Philosophy, and Computing Profession & Ethics

Topic Covered: Professional Ethics, Philosophy of Ethics, Computing Profession, Computing Professions

Introduction

Society

A society is an association of people organized under a system of rules designed to advance the good of its members over time.

Morality

Every society has rules of conduct describing what people ought and ought not to do in various situations. We call these rules morality.

Ethics

Ethics is the philosophical study of morality, a rational examination of people's moral beliefs and behavior.

Ethics is the study of right and wrong, good and bad behavior in human conduct, especially in a professional or societal context.

It answers questions like:

- Should a software engineer insert a backdoor in a program?
- Is it ethical to monitor users without their consent?

Ethics is not the same as law. Something can be legal but unethical — for example, creating addictive app algorithms may be legal but ethically questionable.

Categories of Ethics

Ethics can be broadly categorized into four main types: Descriptive Ethics, Meta-ethics, Normative Ethics, and Applied Ethics.

1. Descriptive Ethics: This branch focuses on describing and observing ethical behavior and beliefs as they exist in the real world. It aims to understand what people do consider morally right or wrong, rather than what should be considered right or wrong. For example, a study examining the ethical values of different cultures would fall under descriptive ethics.

2. Meta-ethics: Meta-ethics delves into the fundamental nature of morality itself. It explores questions about the meaning of ethical terms like "good," "bad," "right," and "wrong". Meta-ethics also investigates the source of moral judgments, whether they are objective or subjective, and how moral knowledge is possible.

3. Normative or Prescriptive Ethics: This branch is concerned with establishing moral standards and principles for determining what is right and wrong. It seeks to provide guidelines for how people ought to behave. Normative ethics includes theories like Consequentialism, Deontology, Virtue Ethics.

4. Applied Ethics: This branch applies ethical theories and principles to specific real-world issues and situations. Examples include: Medical Ethics, Business Ethics, Environmental Ethics, etc.

Theories of Ethics

There are many theories on ethics, but we'll discuss a few important ones.

Relativism

Relativism is the theory that there are no universal moral norms of right and wrong. According to this theory, different individuals or groups of people can have completely opposite views of a



moral problem, and both can be right. Two particular kinds of relativism we'll discuss are subjective relativism and cultural relativism.

Subjective relativism holds that each person decides right and wrong for himself or herself. This notion is captured in the popular expression, "What's right for you may not be right for me."

Cultural relativism is the ethical theory that the meaning of "right" and "wrong" rests with a society's actual moral guidelines. These guidelines vary from place to place and from time to time.

The Divine Command Theory

The divine command theory is based on the idea that good actions are those aligned with the will of God and bad actions are those contrary to the will of God. Since the holy books contain God's directions, we can use the holy books as moral decision-making guides.

The Ethical Egoism

In sharp contrast to the divine command theory, which promotes a concern for others with scriptural injunctions such as "Love your neighbor as yourself," ethical egoism is the philosophy that each person should focus exclusively on his or her self-interest. In other words, according to ethical egoism, the morally right action for a person to take in a particular situation is the action that will provide that person with the maximum long-term benefit.

Ethical egoism does not prohibit acting to help someone else, but assisting another is the right thing to do if and only if it is in the helper's own long-term best interest.

Kantianism

Kantianism is the name given to the ethical theory of the German philosopher Immanuel Kant (1724–1804). Kant believed that people's actions ought to be guided by moral laws, and that these moral laws were universal. He held that in order to apply to all rational beings, any supreme principle of morality must itself be based on reason. Kant gave two categorical imperatives:

First Formulation: Act only from moral rules that you can at the same time will to be universal moral laws.

Second Formulation: Act so that you always treat both yourself and other people as ends in themselves, and never only as a means to an end.



FIGURE 2.5 The second formulation of the Categorical Imperative states that it is wrong for one person to use himself or another person solely as a means to an end.

Utilitarianism (Consequentialism)

The English philosophers Jeremy Bentham (1748–1832) and John Stuart Mill (1806-1873) proposed a theory that is in sharp contrast to Kantianism. According to Bentham and Mill, an action is good if its benefits exceed its harms, and an action is bad if its harms exceed its benefits. Their ethical theory, called utilitarianism, is based upon the principle of utility, also called the Greatest Happiness Principle.

Utility is the tendency of an object to produce happiness or prevent unhappiness for an individual or a community. Depending on the circumstances, you may think of “happiness” as an advantage, benefit, good, or pleasure, and “unhappiness” as a disadvantage, cost, evil, or pain.

Principle of Utility (Greatest Happiness Principle): An action is right (or wrong) to the extent that it increases (or decreases) the total happiness of the affected parties.

We call utilitarianism a **consequentialist theory**, because the focus is on the consequences of an action.

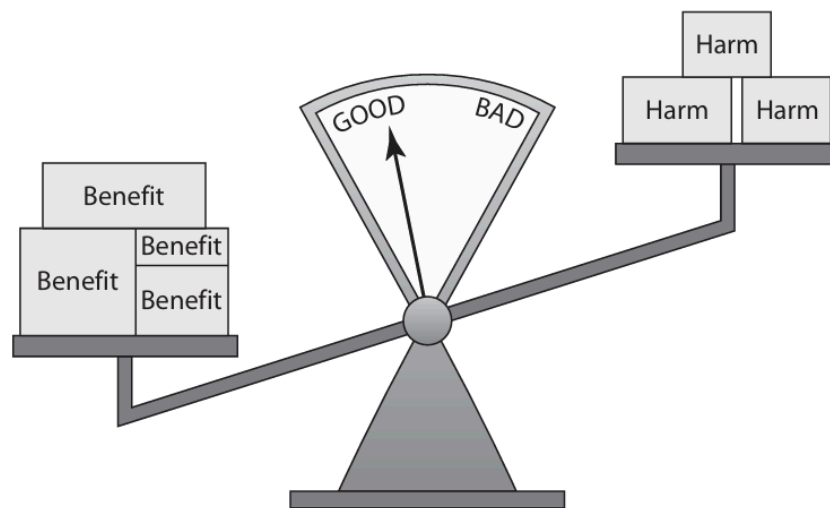


FIGURE 2.6 Utilitarianism is based on the principle of utility, which states that an action is good (or bad) to the extent that it increases (or decreases) the total happiness of the affected parties.

Social Contract Theory

Morality consists of the set of rules governing how people are to treat one another, that rational people will agree to accept, for their mutual benefit, on the condition that others follow those rules as well.

Virtue Ethics

A right action is an action that a virtuous person, acting in character, would do in the same circumstances. A virtuous person is a person who possesses and lives out the virtues. The virtues are those character traits human beings need in order to flourish and be truly happy.

A **vice** is a character trait that prevents a human being from flourishing or being truly happy. Vices, then, are the opposite of virtues.

Deontology

The theory of deontological reason does not concern itself with the consequences of the action but rather with the will of the action. An action is good or bad depending on the will inherent in it. According to deontological theory, an act is considered good if the individual committing it had a good reason to do so. This theory has a duty attached to it. The word “deontology” comes from two Greek words: deon, meaning duty, and logos, meaning science.



For example, we know that killing is bad, but if an armed intruder enters your house and you kill him or her, your action is good, according to deontologists. You did it because you had a duty to protect your family and property.

Professional & Computing Ethics

A **profession** is a vocation that requires a high level of education and practical experience in the field.

Professional ethics refers to the **moral principles and standards of conduct** that govern behavior within a specific **profession**. These ethics ensure that professionals act responsibly, with integrity, and in the **best interest of their clients, colleagues, and society**.

The **computing profession** involves the study, design, development, and application of computer systems and software to solve real-world problems. It is one of the most impactful fields in modern society, influencing every sector, including education, health, business, and government.

Computer ethics, as defined by **James H. Moore**, is the **analysis of the nature and social impact** of computer technology and the **formulation of ethical policies** for its responsible use.

Structure of the Engineering Profession

Two-Tier Structure:

1. First Tier – Engineering Institutions (Chartered Professional Bodies):

- **Definition:**
Independent institutions, each covering one or more engineering disciplines.
- **Examples:**
 - Institution of Electrical Engineers (IEE)
 - Institution of Mechanical Engineers
 - British Computer Society (BCS)
- **Roles & Characteristics:**

- Set **membership rules** and **codes of practice and conduct**.
- Represent **discipline-specific interests** of engineers.
- Provide **expert advice to government**.
- Promote:
 - **Knowledge advancement**
 - **Improvement of standards**
 - **Education in their field**
- **BCS Specifics:**
 - Natural institution for **software engineers**.
 - IEE also admits software engineers, especially in **safety-critical** and **control systems**.
 - **BCS + IEE Collaboration:** e.g., publication of *IEE Proceedings-Software*.

2. Second Tier – Engineering Council (Umbrella Chartered Body):

- **Functions:**
 1. Recognizes institutions as **“Nominated Bodies”** (meets required standards).
 2. **Represents** engineering profession **as a whole**.
 3. **Advises** on matters affecting the **entire profession**.
 4. Maintains the **national register** of qualified engineers and technicians via the **Board for Engineers’ Registration**.

Ethical Guidelines for Computer Professionals

Special Aspects of Professional Ethics

Professional ethics differ from general ethics due to the **unique role and expertise** of professionals. Because clients and the public often **lack technical knowledge**, professionals like computer scientists or doctors have a **special responsibility** to act with honesty, skill, and care.

Their work can **profoundly impact society**, including people's **health, safety, finances, and privacy**, even those who aren't direct clients. As a result, professionals must consider and reduce **risks** related to privacy, security, safety, and usability, and ensure they act in the **best interest of both clients and the public**.

Professional Codes of Ethics

Professional codes of ethics are **formal guidelines** developed by organizations to promote **ethical behavior** within a profession. They serve to:

- Reinforce core **values** like **honesty, fairness, and responsibility**
- Provide **clear guidance** to new or inexperienced professionals
- Help professionals handle **complex ethical situations**

Guidelines and Professional Responsibilities

This section outlines essential **ethical and professional responsibilities** for computer professionals, especially software developers and system designers. The goal is to ensure that systems are **safe, reliable, user-centered, and ethically developed**.

Key Principles and Guidelines

◆ 1. Understand What Success Means

- A system must fulfill its **real-world purpose**, not just perform technically correct operations.

- Developers must see success from the **user's and organization's perspective**, not just the code level.

♦ 2. Include Users in Design and Testing

- Involve **real users** (e.g., medical staff, technicians) to ensure the system is practical and safe.
- Systems fail when developers overlook **actual user needs**.

♦ 3. Plan Thoroughly

- Allocate proper **time and resources** for testing, documentation, and user training.
- Poor planning leads to **ethical compromises** under pressure.

♦ 4. Design for Real Users

- Anticipate **human errors**, typos, confusion, and lack of training.
- Interfaces must be **clear, intuitive**, and include **error-checking**.

♦ 5. Don't Assume Existing Software is Safe

- Software reused from another system may not meet the **safety and reliability** needs of the current application.
- **Re-evaluate** even trusted software when used in new contexts.

♦ 6. Be Honest About Capabilities and Limitations

- Don't mislead clients or users about what the software can do.
- Report **known flaws**, and **take responsibility** for damage caused.



♦ 7. Require Proof of Safety

- In **safety-critical systems**, do not proceed unless there's a **convincing case for safety** — not just absence of known danger.

♦ 8. Set Ethical Default Settings

- Users often **do not change** default settings.
- Defaults should prioritize **privacy, security, and ethical outcomes** when appropriate.

♦ 9. Develop Communication Skills

- Professionals must **effectively explain technical concepts** to clients and coworkers.
- Clear communication ensures users are properly **informed and empowered**.

Computing Professions

- Software Engineer
- System Analyst
- Network Administrator
- Database Administrator
- Cybersecurity Analyst
- AI/ML Engineer
- Web Developer
- IT Consultant

Lecture 2: The Structure of Organizations & Anatomy of Software Houses

Topics Covered: The Structure of Organizations, The Anatomy of Software Houses

The Structure of Organizations

An organization's structure defines **how tasks are divided, coordinated, and supervised** to achieve its goals. Organizational structure plays a critical role in shaping **ethical behavior** and **professional responsibility** within a company. It directly affects communication, decision-making, efficiency, and employee roles.

It answers:

- Who reports to whom?
- Who makes decisions?
- How is work divided and coordinated?

Legal Forms of Organization

In any civilized society, the law **recognizes individuals** (human beings) as entities capable of responsibility, and therefore capable of being held accountable—whether through **contracts, lawsuits, or criminal law**. However, for business activities beyond the smallest scale, it is beneficial to **establish a separate legal identity** through **incorporation**. Below are the primary legal forms of organizations:

1. Sole Trader

- **Definition:** A single individual conducting business under their own name.
- **Legal Status: Unincorporated** (no separate legal identity).
- **Liability:** The **individual is personally liable** for all business debts (including personal assets like a family home).

- **Advantages:** Simple and easy to start; minimal formalities.
- **Risks:** Unlimited personal liability.

2. Partnership

- **Definition:** A relationship between **two or more individuals** carrying on business **with a view to profit** (as per the **Partnership Act 1890**).
- **Legal Status:** **Unincorporated**, though recognized by law.
- **Liability:** Partners are **jointly and personally liable** for the debts and actions of the partnership—even for debts caused by another partner's overreach.
- **Common Usage:** Frequently used in professions like **law** and **medicine** (e.g., solicitors, general practitioners).
- **Drawbacks:**
 - Difficulties in handling **frequent changes** in personnel.
 - Unlimited liability deters broader usage in commercial business.

3. Incorporated Companies

- **Definition:** Organizations that are granted a **separate legal identity**.
- **Incorporation Methods:**
 - **Royal Charter:** Granted by the Privy Council for institutions acting in the **public interest** (e.g., universities, professional bodies).
 - **Statutory Body:** Created via an **Act of Parliament** (e.g., District Health Authorities).
 - **Limited Company** (most common): Businesses incorporated under the **Companies Act** with **limited liability** protection.
- **Advantages:**

- **Separate legal personality**
- **Limited liability** for shareholders
- Easier to raise capital and scale operations

Types of Companies

1. By Public Offering:

- **Public Company (PLC):** May offer shares to the public, Name ends in “Public Limited Company” / “PLC”. Requires issued share capital > £50,000. Heavily regulated.
- **Private Company (Ltd):** Cannot offer shares to the public, Name ends in “Limited” / “Ltd”.

2. By Liability:

- **Unlimited Company:** Shareholders personally liable for all debts (rare).
- **Limited Company:**
 - **By Shares:** Liability limited to owned shares.
 - **By Guarantee:** Members pay a fixed amount on winding up. Used by charities and professional bodies.

Ownership and Membership

● Shares:

Represent ownership; can be bought/sold. Restrictions may apply (especially in small companies).

- **Shareholders (Members):**

Can be individuals or companies. Companies can now have a **single member** (1992 regulation). Special provisions exist for separation of company and sole member.

Constitution of a Company

Every company must possess a **written constitution**, composed of:

1. **Memorandum of Association** – Governs **external relations**.
2. **Articles of Association** – Dictates the **internal management**.
3. **Shareholders' Agreement** – (Optional) Ensures protection for **minority shareholders**.

Memorandum of Association

This document includes:

- **Company Name:** Must be unique and free of restricted terms unless permission is granted (e.g., "Parliament", "Wales").
- **Registered Office:** Stating whether it's in England & Wales, Wales alone, or Scotland.
- **Objects Clause:** Defines the business scope. Since the **Companies Act 1989**, companies can state their object simply as conducting any general commercial business, reducing the risk of *ultra vires* actions.
- **Liability Clause:** States limited liability of shareholders.
- **Authorized Share Capital:** Includes details on the nominal value and number of shares.
- **Declaration of Association:** A formal agreement by initial subscribers to form the company and purchase shares.

Articles of Association

These detail the rules for internal management. Companies can adopt:

- **Table A:** A standard model (amended over time since 1948).
- **Long Form:** Full rules written out.
- **Short Form:** Only deviations from Table A are specified.

Key contents include:

- **Share Allotment Rules**
- **Share Transfer Restrictions:** Especially in private companies to prevent unwanted ownership transfers.
- **General Meetings:** Includes rules for AGMs, EGMs, quorum, proxies, and resolution procedures.
- **Director Appointment and Removal:** Elections, retirements, reappointments.
- **Director Powers:** Board authorized to manage daily operations; some powers may be reserved for general meetings.
- **Dividend Policies and Reserves**

Shareholders' Agreement

To protect minority shareholders:

- It is a **private contract** among shareholders.
- More rigid than Articles; changes require **unanimous consent** of signatories.
- It complements the Articles by outlining binding internal governance commitments.

Directors and the Company Secretary

- **Directors' Role and Responsibilities:**
 - Elected by shareholders to manage the company.

- Hold significant power with limited shareholder control in large companies.
 - Must act **in good faith**, for the **company's best interest**, and with **due diligence**.
 - Must **avoid conflicts of interest** and **disclose personal interests** in company contracts.
 - May be **held personally liable** for company debts if they act negligently.
- **Company Secretary:**
 - Responsible for maintaining statutory records and submitting returns to **Companies House**.
 - Often a qualified professional; may also be a director in small companies.
 - Plays a key role in legal compliance and documentation.
- **Executive vs. Non-Executive Directors:**
 - **Executive Directors:** Involved in daily management.
 - **Non-Executive Directors:** Advisory role but share equal legal responsibility.

Key Elements of Organizational Structure:

Element	Description
Hierarchy	Levels of management from top (executives) to bottom (staff)
Division of Labor	How work is split among individuals and departments
Span of Control	Number of employees managed by a single supervisor
Centralization	Where decision-making authority lies (central vs. distributed)
Formalization	The extent to which rules, procedures, and roles are written and enforced
Departmentalization	How employees are grouped (e.g., by function, product, region)

Common Types of Organizational Structure:

Type	Description
Functional Structure	Organized by departments (e.g., HR, IT, Sales)
Divisional Structure	Organized by product, service, or geography
Matrix Structure	Employees report to both functional and project managers
Flat Structure	Few or no levels of middle management; encourages quick communication
Hierarchical Structure	Clear chain of command from top to bottom
Team-Based Structure	Organized around collaborative project teams

Importance of Organizational Structure

- Clarifies **roles and responsibilities**
- Improves **communication and coordination**
- Affects **efficiency, agility**, and **decision-making speed**
- Supports **goal alignment** and accountability
- Determines how quickly an organization can **adapt to change**



Why Ethics Must Be Built into the Structure

Without ethical alignment, even a well-structured organization can:

- Enable unethical decisions from the top to down
- Allow cover-ups due to poor oversight
- Fail to protect whistleblowers
- Promote profit over integrity

Factors Influencing Job Performance

1. Understanding of Job Requirements
2. Individual Ability
3. Quality of Job Facilities
4. Motivation (Desire to Perform Well)
5. Colleagues' Attitude

Case Study: Demotivation in a Software Team

- **Ambiguous Specifications:** Team members told to “do the best you can.”
- **Poor Role Assignment:** Skill mismatches based on qualifications.
- **Inadequate Resources:** 5 terminals for 25 staff; faulty equipment.
- **Unsuitable Environment:** Converted hangar, focus on “flexitime.”
- **Low Morale:** Frequent job searches; leaving parties common.
- **Lack of Purpose:** No interest in project goals.

- **High Turnover:** Talented individuals leave first.
- **Decline in Performance:** New recruits lose motivation within months.

Root Cause: Mismanagement of understanding, ability, and facility factors leads to low motivation and team decline.

Key Motivators (Assuming Basic Needs Met)

- **Self-Esteem:** Doing meaningful and quality work.
- **Esteem of Others:** Recognition from peers, superiors, customers.
- **Social Satisfaction:** Sense of group belonging.
- **Security:** Stability and confidence in position.
- **Financial Rewards:** Often low on individual priority lists.

Managerial Actions to Motivate

- Discover individual motivators.
- Align performance with improvement in motivators.
- Offer **non-financial rewards**:
 - Consultation and valuing input.
 - Improved working conditions.

Limitations of Financial Rewards

- **May Not Motivate Effectively**, but:
 - **Can Demotivate** if:

- **Inequitable Raises:** Rewards go to complainers, not performers.
- **Recruitment Discrepancies:** New hires paid more than current staff.

Performance Appraisal

Performance appraisal is a critical management activity designed to evaluate staff effectiveness and improve organizational productivity. The foundation of this approach lies in the concept of **Management by Objectives (MBO)**—a management style popularized in the 1970s. Although the term has declined in use, its principles remain central to performance management practices across diverse domains, including software engineering and education.

The core principle of performance appraisal under MBO is the **systematic evaluation of an individual's work performance against previously agreed objectives**. This process typically unfolds over a fixed period (commonly six or twelve months) and involves the following cycle:

1. **Objective Setting:**

Managers and subordinates collaboratively define realistic, achievable, and desirable job objectives. This step requires transparent communication, where managers explain broader organizational goals and identify potential barriers to success.

2. **Performance Evaluation:**

At the end of the set period, the subordinate's performance is reviewed against the agreed-upon objectives. New objectives are also established based on the outcome of this evaluation.

To ensure the success of this process, several key considerations must be observed:

- **Joint Participation in Objective Setting:**

Objectives must be mutually agreed upon to foster ownership and commitment. This collaborative approach ensures feasibility and alignment with both individual capabilities and organizational needs.

- **Verifiability and Balance:**

Objectives should be measurable and clearly defined (e.g., "submit monthly sales reports by the third working day"). However, care must be taken not to prioritize quantifiable tasks over equally essential but non-measurable responsibilities, such as team morale or



professional growth.

- **Role Suitability:**

The appraisal system assumes job consistency. For roles involving highly dynamic or reactive tasks (e.g., communications consultants or support staff), setting fixed long-term objectives may be impractical. In such cases, broader goals like maintaining up-to-date knowledge and ensuring client satisfaction still provide valuable direction.

- **Sustained Management Commitment:**

Consistent and timely performance reviews are necessary for MBO to be effective. In the software industry, performance appraisals are often delayed or overlooked due to pressing operational crises, diminishing their value and impact.

- **Diagnostic, Not Punitive Reviews:**

Reviews should aim to uncover underlying reasons for unmet goals rather than penalize staff. This diagnostic approach allows for course correction and improvement in future performance cycles.

- **Separation from Salary Review:**

While performance may inform salary adjustments, appraisals should not be tightly coupled with compensation decisions. Linking them too closely may discourage honest dialogue during reviews. Moreover, salary changes depend on additional factors such as market demand, organizational equity, and financial viability. Recognizing potential and long-term contribution often carries more weight than short-term goal achievement alone.

Sub-optimization

Sub-optimization occurs when the pursuit of optimal performance in a specific unit or division leads to negative consequences for the overall organization. This issue arises from a fundamental conflict: **optimizing individual components does not necessarily optimize the system as a whole.**

Example 1: The Software Division

Consider a software company division that initially enjoys strong customer satisfaction and high business volume but suffers from poor profitability (around 1% of turnover). A new manager is tasked with improving profitability to 10% within a year. He accomplishes this by:

- Raising service charges significantly
- Suppressing salary increases

While the financial target is met, the side effects are damaging:

- **Turnover drops by 20%**
- **Staff numbers reduce by 30%**
- **Long-standing clients leave**
- **Top-performing employees resign**
- **The division's reputation suffers**

Despite achieving his stated goal, the manager's actions undermine long-term success. He is promoted, leaving behind a destabilized division—highlighting a classic case of **short-term objective fulfillment at the expense of organizational sustainability**.

Example 2: The XYZ Group

XYZ is a group of autonomous subsidiaries, including **XYZ Tramways Ltd** and **XYZ SuperRail Ltd**. When the government of Pontevedro announces a major transport integration project, both subsidiaries bid separately:

- XYZ Tramways partners with ABC Ferrovie.
- XYZ SuperRail partners with PQR Strassenbahnen.

Both bids are shortlisted due to high quality and reputation. In the final stage, **XYZ Tramways undercuts its own price by 20%**, winning the contract. This results in:

- **Internal competition between sister companies**
- **Loss of potential group synergy**

- **Reduced profit margin due to price undercutting**

While the group won the project, it **lost potential profitability** and risked internal conflicts—symptomatic of **lack of inter-unit collaboration and poor strategic alignment**.

Lessons and Mitigation Strategies

1. **Define Objectives Holistically:**

In the software division case, the problem stemmed from **incomplete and narrowly defined objectives**. A better directive would be:

"Raise profitability to 10% **while maintaining or improving current sales and staffing levels**."

Like writing software requirements, one must avoid assumptions and clearly articulate all success criteria to prevent unintended outcomes.

2. **Encourage Strategic Collaboration:**

XYZ's failure reveals the risks of **excessive decentralization**. Independent units should be encouraged to **share information, align goals, and coordinate efforts**. Group policy must support:

- Informal and formal inter-subsidiary meetings
- Joint ventures where logical
- Open communication between senior management teams

3. **Avoid Internal Competition That Harms the Whole:**

While some competition can drive innovation, **unregulated internal rivalry** (such as price wars between subsidiaries) can erode group value.

4. **Promote a Culture of Transparency and Cooperation:**

Organizational culture plays a key role. In collaborative environments, managers are

more likely to share plans, align objectives, and mitigate conflict. In contrast, siloed organizations encourage secretiveness and fragmented goals.

Anatomy of a Software House

A **software house** is a company that specializes in developing, maintaining, and delivering **software solutions** to clients or consumers. It can vary in size and specialization but generally follows a structured model that includes key departments, workflows, and roles. Understanding the **ethical structure** of a software house is essential, as each department holds specific **ethical responsibilities** that directly affect users, clients, and society. The ethical framework must be embedded not just in the code, but in the **culture, workflow, and decisions** across the organization.

Departments of a Software House

Department/Unit	Key Roles & Functions
Project Management	Planning, scheduling, resource allocation, client communication
Software Development	Writing code, testing features, implementing requirements
Quality Assurance (QA)	Testing software for bugs, usability, and performance issues
UI/UX Design	Designing user interfaces and optimizing user experience
DevOps / IT Support	Automating deployments, maintaining servers, system stability
Business Analysis	Understanding client needs and writing functional specifications
Sales & Marketing	Client acquisition, branding, and product promotion
Human Resources (HR)	Hiring, training, employee welfare, and compliance

Finance & Admin	Budgeting, accounting, and office management
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Common Workflow in a Software House

1. **Client Interaction / Requirement Gathering**
 - Handled by Business Analysts & Project Managers
2. **Planning & Documentation**
 - SRS, SDLC model selection (Agile, Scrum, Waterfall)
3. **Design Phase**
 - UI/UX team creates wireframes, mockups, and prototypes
4. **Development Phase**
 - Frontend and backend developers build the system
5. **Testing Phase**
 - QA team runs manual and automated tests
6. **Deployment & Maintenance**
 - DevOps manages deployment; the support team handles post-launch issues

Typical Job Roles

Role	Description
Frontend Developer	Works on user-facing features (HTML, CSS, JavaScript, etc.)
Backend Developer	Handles server-side logic, databases, and APIs
Full Stack Developer	Manages both frontend and backend
QA Engineer	Tests and validates software functionality

Project Manager (PM)	Oversees progress, ensures timelines, and client satisfaction
UI/UX Designer	Designs interfaces and optimizes interaction flow
DevOps Engineer	Automates deployment, manages cloud infrastructure
Technical Writer	Writes documentation, manuals, and user guides

Ethical and Professional Responsibilities

- Ensure **data privacy**, **intellectual property protection**, and **client confidentiality**
- Avoid plagiarism, code theft, and misleading product claims
- Respect **workplace diversity**, **equal opportunity**, and **fair treatment**
- Maintain **transparency** in timelines, costs, and system capabilities
- Address **security risks** and test rigorously before release

Organizational Structure

CEO / Founder

└─ Technical Director / CTO

| └─ Development Team

| └─ QA / Testing

└─ Project Management

└─ Business Analysis

└─ UI/UX & Design

└─ DevOps & Infrastructure

└─ Sales, HR, Finance

Ethical and Professional Principles

Principle	Meaning in Practice
Honesty	Disclose limitations, bugs, and realistic deadlines to clients/users
Privacy	Ensure all user data is securely stored and not misused
Accountability	Accept responsibility for code errors or product failures
Non-maleficence	Do not create software that causes harm (malware, biased algorithms, etc.)
Fairness & Inclusion	Ensure accessibility for all users; promote diversity in hiring
Transparency	Openly communicate with clients about risks, cost, and scope

Common Ethical Issues in Software Houses

- Launching software without adequate testing (safety risks)
- Ignoring user privacy laws (e.g., GDPR, PECA)
- Reusing open-source code without attribution (IP violation)
- Designing manipulative interfaces (e.g., forced subscriptions)
- Prioritizing speed/profit over quality and security

Monitoring Financial Performance

- 
- **Performance Monitoring:** Involves comparing monthly income and expenditure against the budget to identify deviations and take corrective action.
 - **Practical Difficulties:** Include random fluctuations due to:
 - **Staffing Variance:** Budget assumes a linear net monthly increase in staff (e.g., +2.5/month), but real figures vary monthly.
 - **Large Project Effects:**
 - Proposal preparation can divert senior staff from revenue-generating work.
 - After a large project ends, staff may become available faster than redeployment rate.

Lecture 3: Finance and Accounting

Topics Covered: Finance and Accounting

Introduction

Finance and accounting are critical functions within a software or technology organization. Understanding these concepts helps computing professionals — especially software engineers, managers, or startup founders — to manage **budgets, capital, pricing, and audits** responsibly and ethically. Any organization cannot be successful unless its finances are soundly managed.

According to **Quinn** and **Baase**, financial literacy is part of being an **ethically responsible professional**, especially when decisions affect clients, investments, and long-term sustainability.

What is Finance?

Finance is the **management of money**, including how it is **raised, invested, and allocated**. It focuses on planning the organization's **financial future**, analyzing risks, and making strategic decisions.

Key Areas of Financing

Area	Description
1. Financial Planning	Setting short-term and long-term financial goals and developing strategies to achieve them.
2. Capital Management	Managing sources of funds (e.g., equity, loans) and ensuring efficient use of capital.
3. Investment Decisions	Analyzing opportunities for investment in assets, R&D, or expansion to maximize returns.
4. Risk Management	Identifying and minimizing financial risks (market risk, credit risk, operational risk).

5. Cash Flow Management	Monitoring inflow and outflow of cash to ensure liquidity and business continuity.
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What is Accounting?

Accounting is the process of **recording, summarizing, analyzing, and reporting** financial transactions. It ensures **financial transparency** and helps stakeholders understand how the organization is performing.

Key Areas of Accounting

Area	Description
1. Bookkeeping	Daily recording of all financial transactions (sales, expenses, payments).
2. Financial Reporting	Preparing financial statements (balance sheet, income statement, cash flow).
3. Cost Accounting	Tracking and analyzing costs to improve efficiency and determine product pricing.
4. Tax Accounting	Ensuring compliance with tax laws and calculating correct tax liabilities.
5. Audit and Assurance	Verifying the accuracy of financial records and ensuring regulatory compliance.



Key Issues

1. The Need for Capital

Contract hire services: Offering services as programmers on a daily-fee basis.

Market gap: Opportunity perceived for new software packages not currently available.

Invoice delay: Payment delays range from 1 to 12 months post-invoicing, requiring initial financial support.

Startup capital needs:

- **Living expenses:** For at least 3 months before payment cycles begin.
- **Operational expenses:**
 - Salaries (for founders and staff)
 - Rent, utilities (heating, lighting)
 - Equipment and consumables
 - Advertising and marketing
 - Miscellaneous (stationery, travel, etc.)
 - Interest on borrowed money

Revenue lag: Even after product development, sales take time to meet ongoing costs.

Business Plan:

- **Purpose:** To attract and convince potential funders.

- **Contents:**

- Company description and technical feasibility
- Founders' expertise
- Market size and competition analysis
- Financial performance forecast

2. Sources of Funds – Structured Summary

Funding options are categorized into:

1. Grants

- **Grant:** Non-repayable sum of money provided (mainly by government or EU).
- **Conditions:**
 - Must be used for stated purposes.
 - Often only a **percentage** of total development cost.
 - Require **co-funding** from other sources.
- **Factors Affecting Availability:**
 - Geographic location of the company.
 - Number of employees.
 - Current government policy.
- **Example Assistance:**

- Free or discounted premises for 12 months.
- £15,000–£20,000 grant for hiring 5–6 people; similar grant when staff reaches 10–12.
- **Limitations:**
 - Often not suitable for software companies (low physical capital investment).
- **Special Programs:**
 - **SMART (UK)** and **Framework V (EU)**.
 - May require **collaborative projects** with firms in different EU countries.
 - Typically limited to **50%** (or less) of development cost.

2. Loans

- **Loan:** Borrowed money that must be repaid with **interest**.
- **Security (Collateral):**
 - Often required; may involve company or **directors' personal assets**.
- **Loan Types:**
 - **Overdraft:**
 - Flexible bank facility allowing spending beyond account balance.
 - Interest on used amount only.
 - Can be **withdrawn anytime**, risky.
 - **Short-Term Loan:** Not detailed, but implied for short-term needs.

- **Long-Term Loan:**
 - Fixed period, fixed interest.
 - Capital repaid at end; lender **cannot withdraw** early if payments made on time.
- **Soft Loan:**
 - Special government-supported loan.
 - Lower interest, **no security required**.
 - Available mainly for **startups**.

3. Equity Capital & Gearing

- **Equity Capital:**
 - Funds raised by **selling ownership shares** in the company.
- **Gearing:**
 - Ratio of loan capital to equity capital.
 - **High Gearing Risks:**
 - High loan interest may consume profits.
 - Shareholders may receive nothing in bad years.
 - Encourages **risky behavior** (low risk to shareholder, high potential gain).
- **Investor Returns:**
 - Lenders: Fixed interest only.

- Shareholders: Potentially high returns **if** company succeeds (and higher risk of loss).

3. Budgeting and Monitoring

Definition of Budget

- A **budget** is a **financial forecast** covering the current or upcoming financial year.
- Reflects the **financial part of a business plan**, supported by:
 - **Marketing analysis**
 - **Technical feasibility**

Scope of Budget

- Should include predictions for **all annual financial statements**
- In practice, **managers focus mostly on income and expenditure budgets**

Income Forecasting

Based on different **sources of revenue**: **Existing contracts**, **New contracts from current clients**, **Contracts from new clients** (most uncertain)

Budgeting Process

Iterative in nature: Initial budget usually shows **excess expenditure over income**. Adjustments are made until:

Sales > Expenditure, **Operational feasibility** is assured, **Sales targets** are realistic

4. Sales and Order Intake

Sales Monitoring:

A crucial managerial activity as a company's **costs, pricing, and overhead** depend largely on sales levels.

Product Categories Based on Sales Nature:

- **Mass-produced goods** (e.g., chocolate bars, floppy disks):
Produced at a **steady rate**, stored in inventory, and dispatched **immediately** after order—leading to **short payment cycles**.
- **Produce-to-order model** (e.g., custom software):
Products or services are **customized** for each order, often leading to **longer payment cycles** and requiring **clear tracking of order intake**.

Order Intake vs. Sales:

- **Sales:**
Total value of **invoices issued** during a month.
- **Order Intake:**
Total value of **orders received**, regardless of delivery or invoicing.

Revenue Forecasting from Orders:

- Each order must be analyzed to **predict future monthly sales**.
- **Revisions** are required as **terms change**, such as manning adjustments or inflation-based increases.

5. Costing

Understanding the **cost structure** and **financial position** of a company is fundamental to strategic pricing, investment decisions, and overall financial health. Costing focuses on how much it takes to produce a product or deliver a service.

Cost influences pricing and competitiveness. Costing can be segmented into four key categories:

1. Raw Materials & Bought-in Items

- **Raw materials:** Transformed during production (e.g., steel).
- **Bought-in items:** Used as-is in the final product (e.g., chips).

2. Equipment Costs

- **Depreciation:**

- *Straight-line method:* Equal annual reduction.
- *Reducing-balance method:* Percentage reduction each year.

- **Interest & Opportunity Cost:**

- Applicable if equipment is funded via loans or capital investment.

- **Fixed Costs:** Do not vary with usage (e.g., insurance).

- **Variable Costs:** Depend on use (e.g., maintenance).

- **Marginal Cost:** Additional cost of producing one more unit.

3. Labour Costs

- Includes **salary, employer contributions, and benefits.**
- **Effective hourly rate** = Total annual cost ÷ Actual productive hours.

4. Overheads

- **Indirect costs** (e.g., admin, management, facilities).
- Categorized into:
 - *Departmental overheads* (linked to specific teams).
 - *Corporate overheads* (general company-wide expenses).
- Allocation Methods:
 - *Input-based:* As a percentage of labour or machine hours.

- *Output-based:* As a mark-up on products or services.
- Key Principles:
 - Ensure consistency and fairness in allocation.
 - Avoid sub-optimization—support overall company efficiency.
 - Maintain competitive pricing.

6. Pricing – General Concepts

Profit Constraint:

Prices must ensure that long-term revenues exceed operational costs. Sustained losses are unsustainable.

Single Product Case:

Minimum price is set to cover costs at a given sales volume.

Real-world Limitations:

Due to uncertain demand-price relationships, most companies set prices as:

Price = Cost + Profit % [+ Overhead %] → Adjusted for market

- Discounts for quantity or preferred customers are common.

Pricing in Software Companies

1. Pricing Service-Based Contracts (e.g., Consultancy, Programming)

- **Base:** Labour cost + Overheads
- **Adjustments Based on:**
 - **Staff Utilization:** Idle staff may justify lowering price.
 - **Client Value:** Prestige or strategic importance may justify discounting.
 - **Market Rates:** Competitor pricing may push adjustments.

2. Pricing Bespoke (Fixed Price) Contracts

- **Estimation Includes:**
 - Labour, travel, subsistence
 - Bought-in software/hardware (including related costs)
 - Financing cost (due to delayed stage payments)
 - Guarantee/servicing cost
 - Risk margin (up to 50%)
- **Post-Estimation Adjustment:** As influenced by client and market, but **risk makes underpricing dangerous**—has led to financial collapse of major companies.

3. Pricing Software Products

- **Cost Considerations:** Development costs, Maintenance & support (fixed + variable), Selling infrastructure
- **Sales Volume Sensitivity:** Low price + low sales = disaster,

High price = fewer sales, higher margin
- **Risk of Underpricing:** Low prices may signal poor quality or misfit to customer expectations—especially risky in software industry.

7. Working Capital and Cash Flow

Accrual Accounting:

- **Revenue** is recorded when an **invoice is issued**, not when payment is received.
- **Expenditure** is recorded when payment is **authorized**, not when cash is disbursed.

Problem: A company can be **profitable** on paper but still **unable to pay bills** due to timing gaps between revenue recognition and actual cash inflow.

Reasons for Cash Flow Problems Despite Profitability

1. **Invoice Payment Delays:** Verification of goods/services. Batched payments (e.g., monthly). Companies deliberately delaying creditor payments.
2. **Work in Progress (WIP):** Long projects (e.g., bespoke software) incur costs long before payment. **Stage payments** help, but often do **not cover actual cost incurred** up to that point.

Working Capital: Defined as the **cash needed to cover day-to-day operations**, bridging the gap between:

- **Cash outflows** (e.g., wages, materials)
- **Cash inflows** (e.g., delayed payments)

Different from Investment Capital: Investment Capital = used for **long-term productivity enhancement**, like machinery or product R&D.

Cash Flow Prediction: A **12-month forecast** showing: **Cash inflows** (based on invoice payment cycles), **Cash outflows** (e.g., quarterly rent), **Cumulative cash position**

- Helps identify: **Maximum working capital required, Timing of cash gaps**
- In contracting companies, often **revised monthly**.

8. Assessing Investment Proposals

Investment Selection Criteria: Investment resources are **limited**—companies must **choose between competing proposals**. Evaluation factors include:

- **Strategic alignment** (with long-term goals)
- **Risk level**
- **Resource availability**

- **Financial return** — typically assessed using **Discounted Cash Flow (DCF)**.

Discounted Cash Flow (DCF)

- Recognizes that **money now is worth more than the same amount in the future** (time value of money).
- **Discount Factor:**

$$DF = \frac{1}{(1 + r)^t}$$

- r = cost of capital (as a fraction)
 - t = number of years into the future
- **Net Present Value (NPV):**
The sum of all discounted net cash flows over the project's life.

$$NPV = \sum_{i=0}^n \frac{X_i}{(1 + r)^i}$$

where X_i is the net cash flow in year i .

- A **positive NPV** = project is financially viable.

DCF & Inflation: Inflation is **automatically handled** if Monetary cash flows are used with the actual interest rate. OR adjust the cost of capital using:

$$\frac{m - i}{m + i}$$

where:

- m = monetary cost of capital
- i = inflation rate

Decision Guidelines



A project may be **rejected** if:

- NPV < 0
- Payback period > company threshold

When comparing multiple viable projects:

- Prefer **highest NPV** (long-term value)
- Or **shortest payback** (lower risk)

Risk & Sensitivity in DCF

- DCF appears precise but is based on **uncertain forecasts**.
- Sources of uncertainty in software projects: Underestimated development effort, Delays in release, Poor market response, Early competitive products
- **Sensitivity Analysis** is crucial:
 - Run DCF with **different scenarios**.
 - If small changes make the project unviable → **High risk**.
 - If still viable under multiple assumptions → **Lower risk**.

9. Annual Statements

Legal Requirements

- **Companies Act 1985** mandates that **limited companies** prepare and file annual accounts comprising:
 1. **Balance Sheet**



2. Profit and Loss Account

3. Auditor's Report

- **Stock-exchange-listed companies** face even **stricter disclosure rules**.

The Balance Sheet: Snapshot of financial position at one date (usually year-end).

Equation: $\text{Assets} = \text{Liabilities} + \text{Shareholders' Funds}$

Assets:

- **Fixed Assets** (long-term productive capacity): *Tangible* (e.g., buildings, equipment), *Intangible* (e.g., copyrights, goodwill), *Investments*
- **Current Assets** (converted to cash within 12 months): Cash & bank balances, Debtors (unpaid invoices), Stock (finished goods), Work in progress / amounts recoverable on contracts
- **Liabilities:**
 - **Current Liabilities** (due within 12 months): creditors, accrued charges
 - **Long-term Liabilities** (due after 12 months): loans, provisions

Cash Flow Statement

- **Reconciles** changes in **cash** between two balance-sheet dates.
- **Definition of Cash:** Cash + cash equivalents – overdrafts/short-term borrowings.
- **Purpose:** Explains how profits and asset movements translate into cash movements.

10. Capital and Its Maintenance

Concept of Capital Maintenance



Capital maintenance ensures the **original capital contributed by shareholders remains intact**, acting as a **safeguard for creditors**.

Share capital serves as a **guarantee** for obligations; lenders consider its size before offering loans.

Legal & Financial Restrictions

- **Issuing shares at a discount** (below nominal value) is **prohibited**, to prevent capital received from being less than issued capital.
- **Return of original capital** to shareholders is **not allowed**.

11. Auditing – Structured Summary

Legal Requirement

- **Companies Act 1989**: Requires company accounts to include an **auditor's report**.
- **Purpose**: To provide assurance that accounts give a "**true and fair view**" of the company's state.
- Auditors must meet **independence criteria**.

Audit Procedure

- **Not all transactions checked**: Auditors do **not verify every financial transaction**.
- They check:
 - **Financial procedures and controls** (adequacy)
 - **Samples of transactions and assets** (e.g., confirming debtors or verifying physical stock)
 - Use **analytical review** to validate overall figures.

Limits of Responsibility

- **Standards compliance only:** Auditors ensure compliance with **accounting standards**, not whether choices made are the best.
- **Going concern:**
 - Auditors check for viability **only if** they have reasons to suspect insolvency.
 - If suspicious, they assess **cash flow** and **credit assurances**.
 - If not, **no obligation** to investigate further.

Fraud Responsibility

- **Directors** are primarily responsible for fraud prevention and detection.
- **Auditors** must work in a way to **reasonably expect** detection of serious misstatements.
- If fraud is found, auditors must **report to directors**.
- **Internal auditors** in large firms handle routine compliance and fraud prevention.

Auditor Independence (Cadbury Report Highlights)

- Concern: Auditors often build closer ties with **management** than with **shareholders**.
- Risk: Audit firms may also perform **non-audit work**, affecting impartiality.
- **Rejected suggestions:**
 - Banning audit firms from non-audit work.
 - Mandatory rotation of audit firms.
- **Recommendations:**

- 
- Separate disclosure of **audit and non-audit fees**.
 - **Rotation of audit partners** (not the firm).

Ethics in Finance and Accounting

- Avoid fraud and false reporting
- Be transparent with investors and clients
- Ensure fair pricing and honest contracts
- Protect sensitive financial data

Ethical lapses in finance can destroy a company's reputation and lead to legal consequences.

Lecture 4: Introduction to IT Laws & Legal Responsibilities

Topics Covered: Computing professions and Law, Computer Contract

Computing Law

Computing law refers to the **legal frameworks** that govern the **development, use, storage, and dissemination** of information through computer systems. As computing technology evolves, new legal challenges emerge involving **privacy, intellectual property, cybersecurity, digital contracts**, and **freedom of expression**.

Definition and Scope

Computing law encompasses legislation that applies to **the use of digital technology**, including software development, data processing, internet communication, and ethical use of information systems.

Core Areas

Area	Description
Privacy & Data Protection	Laws that govern how personal data is collected, stored, processed, and shared.
Intellectual Property (IP)	Protection of software, digital media, and inventions through copyright, patents, and trade secrets.
Computer Crime & Misuse	Cybercrimes such as hacking, phishing, DDoS attacks, identity theft, and spreading malware.
Cybersecurity Laws	Legal requirements for safeguarding systems against unauthorized access and data breaches.
Digital Contracts & E-Commerce	Legal recognition of digital agreements, terms of service, and consumer rights online.



Freedom of Speech vs. Censorship	Balancing free expression online with regulations to control hate speech, extremism, or misinformation.
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Examples of Computing Laws

Law/Act	Description
GDPR (EU)	Strong data protection and user consent framework.
Computer Fraud and Abuse Act (USA)	Federal law against hacking and unauthorized access.
Digital Millennium Copyright Act (DMCA)	Protects IP in the digital world (e.g., against illegal software sharing).
PECA (Pakistan)	Prevention of Electronic Crimes Act – handles cybercrime, fake identity, data breaches.

Importance of Computing Law

According to **Quinn**, understanding computing law is critical for:

- Protecting users' rights
- Avoiding legal liability
- Creating trustworthy, ethical systems
- Complying with international standards

Computer Contracts

Definition

A **computer contract** outlines the **rights, obligations, responsibilities, timelines, pricing, and legal terms** associated with any computer-related service or product.

It can involve **software development, IT consulting, hardware supply, cloud hosting, support & maintenance, or licensing.**

Fewer Issues in Contract Law compared to Intellectual Property and Criminal Law.

Types of Computer Contracts

Type	Purpose
Software Development Contract	Agreement for building custom software for a client
Software License Agreement	Grants permission to use software under specific terms (e.g., EULA)
Maintenance & Support Contract	Covers post-deployment bug fixes, updates, and technical support
Service Level Agreement (SLA)	Specifies service quality, uptime, and performance for IT services
Cloud Service Contract	Defines data ownership, uptime, and access terms for cloud platforms
IT Consulting Agreement	Engages a firm or individual to advise or manage IT solutions



Outsourcing Contract	Shifts specific IT tasks (e.g., customer support) to a third party
Data Processing Agreement (DPA)	Required when handling personal data (especially under GDPR/PECA)

Key Elements to Include

Deliverables

Not just the software, but also:

- Source code
- Build/install instructions
- User and technical documentation
- Test data and test results

Ownership of Rights

Clearly define:

- What **intellectual property (IP)** is transferred
- Who retains the right to reuse, modify, or resell the software

Confidentiality

Client data and project details must be protected. No sharing with third parties without written permission.

Payment Terms

Example clause: Payment shall be due within 30 days of invoice. Delays beyond this may attract a 2% interest or contract termination.

Penalties and Change Payments

Include:

- **Penalties for late delivery**
- **Compensation for client-requested changes** or rework

Client's Obligations

Clients must:

- Provide accurate input and resources on time
- Cooperate during the testing and review phases

Contracts for the Supply of Custom-Built Software at a Fixed Price

Fixed-price software contracts transfer much of the financial risk of project overruns onto the supplier. To manage that risk—and to ensure clarity about deliverables, rights, and obligations—such contracts must be carefully structured. Below is a detailed breakdown of the typical elements and clauses found in a well-crafted, standard-form fixed-price software agreement.

1. Structure of the Contract

Most software houses adopt a **standard form contract** template, minimizing drafting costs and ensuring consistency across projects. A typical fixed-price agreement comprises:

1. **Introductory Section**
 - Parties' legal names, registered addresses, and authorized signatories
 - Date of execution
 - Definitions of recurring terms (e.g. "Company," "Client," "Acceptance Tests")

2. **Standard Terms & Conditions**

- Clauses governing scope, payment, liability, confidentiality, IP, change control, termination, dispute resolution, etc.
- These provisions remain identical from one engagement to the next.

3. **Project-Specific Annexes (Appendices)**

- Requirements specification (dated and version-numbered)
- Deliverables list
- Project schedule and milestone payment plan
- Staffing and rates table
- Client responsibilities matrix

The **standard terms** reference the annexes for all project-specific data, preserving the master document unchanged.

2. The Introductory Section

● **Contract Composition**

Clearly state that the “Agreement consists of this Introductory Section, the Standard Terms & Conditions, and the Annexes listed below (and only those documents).” This prevents extraneous promises (e.g., off-hand remarks by sales staff) from entering the contract.

● **Definitions**

A concise glossary—either alphabetical or in-sequence—ensures uniform interpretation.

Example: “‘Acceptance Tests’ means the test cases and expected results set out in Annex D, Issue 3, dated 5 March 20XX.”

● **Execution Details**

Parties sign and date; corporate representatives confirm they have authority to bind their organizations.

3. What Is to Be Produced

● **Requirements Specification**

- Unique reference (date, issue number)

- Clear statement that all changes must follow the contract's variation procedure

- **Change-Control Process**

- Formal submission of change requests
- Impact analysis on scope, schedule, and price
- Written change order signed by both parties

4. What Is to Be Delivered

List every tangible and intangible deliverable in an annex, for example:

Deliverable	Description
Source Code	Complete, compilable and commented
Build/Install Scripts	Command files and configuration settings
System Design & Code Documentation	UML diagrams; module interface descriptions
User & Operations Manuals	Step-by-step guides, troubleshooting notes
Maintenance Tools	Custom utilities, test harnesses
Training Materials & Sessions	Slide decks, instructor-led workshops
Test Data & Results	Master test suite and reported outcomes

5. Ownership of Rights

- **Default Rule**

Under most copyright statutes (e.g. UK Copyright, Designs & Patents Act 1988 §11), an employer owns programs written by employees. Independent contractors retain their

copyright unless they assign it by **written** agreement.

- **Assignment vs. Licence**

- **Assignment:** Full transfer of copyright to the Client—typically conditioned on final payment.
- **Exclusive Licence:** Client alone may use the software; copyright remains with the Company.
- **Non-Exclusive Licence:** Company may license others; Client’s licence may include veto rights over sublicense grants (mind competition law constraints).

- **Contractual Clauses**

- “All Intellectual Property Rights in the Deliverables shall, upon payment in full, vest in the Client.”
- Or, “Company hereby grants Client a perpetual, royalty-free, non-exclusive licence to use the Deliverables on Client’s systems worldwide.”

- **Escrow of Source**

To guard against Company insolvency, require deposit of source code and build tools with a third-party escrow agent, releasable upon defined “trigger events” (e.g. bankruptcy, breach).

6. Confidentiality

Mutual nondisclosure obligations protect each party’s trade secrets and sensitive data:

Each Party shall hold in strict confidence all Confidential Information of the other and not disclose it, save to its own employees and advisors on a need-to-know basis, and shall use it solely to perform obligations under this Agreement.

7. Payment Terms

- **Invoicing & Due Dates**

- “Client shall pay each undisputed invoice within 30 days of receipt.”
- Late-payment interest: e.g. “2% per month above Bank Base Rate.”

- **Milestone Schedule**

Milestone	% of Total Price	Due Date
Contract Signature	15%	Upon signing
Design Approval	20%	10 days after design sign-off
Beta Release	30%	5 days after beta delivery
Final Acceptance	25%	10 days after acceptance tests complete
Warranty Expiry	10%	End of 90-day warranty period

8. Calculating Payments for Delays & Changes

- **Client-Caused Delays**

Pre-defined **daily rates** by staff grade enable swift compensation:

“If Client fails to provide required inputs by the agreed date, Company may invoice for additional effort at the rates in Annex B.”

- **Variation Orders**

- Change request reviewed at progress meeting
- Written change order details scope, cost delta, and schedule impact

9. Penalty Clauses

Although rare, a **delay penalty** may appear:

“If Final Acceptance does not occur by the Scheduled Acceptance Date, then Company shall forfeit £X per week of delay (capped at £Y total).”

Providers resist harsh penalties (they increase bid prices and risk disincentivize completion once penalties approach the cap).

10. Client Obligations

To facilitate on-time delivery, the contract should assign the Client clear duties, for example:

Client Deliverable	Due Date
Access to legacy system docs	Within 5 business days of start
Test environment provisioning	10 business days before integration testing
Key-user availability for reviews	At least 3 days' notice before each milestone

11. Standards & Methods of Working

Specify which process framework applies:

- Company's ISO 9001-certified quality procedures
- Client's internal QA standards (e.g. CMMI Level 3)
- Coding standards (e.g. MISRA C, W3C accessibility guidelines)

Clearly state any audits or peer reviews the Client may require.

12. Progress Meetings

Mandate regular status meetings (e.g. weekly):

"The Parties shall hold a Progress Review every Friday at 10 am. Minutes, once countersigned, shall constitute evidence of Milestone completion or agreed Variations."

13. Project Managers



Each side appoints a **Project Manager** with authority to:

- Approve change orders up to £Z
- Schedule milestone tests
- Coordinate resource commitments

List each manager's name, contact details, and delegated financial limits.

14. Acceptance Procedure

- **Test Plan Delivery:** Client provides test cases and pass/fail criteria at least 10 days before testing.
- **Test Execution:** Joint execution—Company runs tests in Client's facility with Client observers.
- **Defect Resolution:** Major defects must be corrected within 5 days; re-test window of 3 days.
- **Acceptance Certificate:** Signed by both PMs, triggering the final milestone payment.

15. Warranty & Maintenance

- **Warranty Period:** e.g. 90 days post-acceptance—"Company will correct any Defect free of charge."
- **Post-Warranty Maintenance:** Optional T&M support:
 - Hourly rates by grade (Annex B)
 - Annual "support retainer" of N days at discounted rate

16. Indemnity

Mutual indemnities against third-party IP claims:

“Company shall indemnify and hold harmless Client from liability arising out of infringement by the Deliverables of any third-party Intellectual Property Right, provided Client gives prompt notice and sole control of the defense.”

17. Termination

- **For Convenience:** Client may terminate on 30 days’ notice; Company paid for work-in-progress plus a redeployment fee equal to 10% of unbilled contract value.
- **For Cause:** Material breach not cured within 15 days—non-breaching party may terminate immediately and recover damages.

Ownership of partially completed work must be addressed (e.g. “Upon termination, Company grants Client a non-exclusive licence to any Deliverables produced to date”).

18. Dispute Resolution & Arbitration

- **Arbitration Clause:**

“Any dispute not resolved within 30 days shall be submitted to arbitration under the Arbitration Act 1996. The arbitrator shall be nominated by the President of the British Computer Society.”

- **Governing Law & Venue:**

“This Agreement shall be governed by the laws of England and Wales, and any arbitration held in London.”

19. Inflation Adjustment

Protect long-running contracts against cost inflation via indexation:

“Prices shall be adjusted annually on 1 April in line with the Business Costs Index published by the ONS, pro-rated for partial periods.”

20. Applicable Law

For cross-border engagements:

“This Agreement shall be governed by and construed in accordance with the laws of England and Wales. The United Nations Convention on Contracts for the International Sale of Goods (CISG) is hereby excluded.”

Other Contracts

In professional software development, understanding different types of software services contracts is essential for both service providers and clients. This section explores three widely used contractual arrangements—**Contract Hire, Time and Materials**, and **Consultancy Contracts**—focusing on their structure, risk allocation, legal responsibilities, and practical implications.

1. Contract Hire

Definition:

A **Contract Hire** agreement is one where the supplier provides personnel to the client, but the supplied staff work **under the client’s direction**.

Key Features:

- Staff are paid based on **daily or hourly rates**.
- Responsibility of the supplier is limited to:
 - Providing **competent personnel**.
 - Replacing staff if they become unavailable or unsuitable.
- **Ownership of IP** (Intellectual Property) must be contractually addressed.
- Commonly known as **“body shopping”**.
- Also used in **freelance agreements** (individuals offer services directly).

Legal and Practical Simplicity:

- Much simpler than fixed-price contracts.
- Issues such as delay penalties, acceptance testing, and liability typically do **not arise**.

- However, it is important to **clearly define the scope and IP rights** to avoid later disputes.

2. Time and Materials (T&M)

Definition:

A **Time and Materials** contract (often called **cost-plus**) is a hybrid model where the client pays based on actual time and resources used, rather than a fixed project cost.

Structure:

- Labour is charged similarly to contract hire.
- **Material costs** (e.g., hardware, software tools) are also reimbursed.
- May include a **maximum payment cap**, beyond which further negotiation is needed.

Advantages:

- **Flexibility**: Ideal for projects where the full scope cannot be defined initially.
- **Cost Efficiency**: Avoids overpayment due to supplier-added contingencies.
- **Client retains control** over how the budget is used.

Risks for Clients:

- No **guarantee of completion** within a specific time or budget.
- **No acceptance testing clauses** with legal weight—though testing may be done informally.

Industry Trends:

- Despite flexibility, there's a **shift towards fixed-price models**, especially in **government and defence** sectors due to budget predictability.

3. Consultancy Contracts

Definition:

A **Consultancy Contract** involves the supplier (consultant) analyzing and advising an organization, typically resulting in a **report or strategic document** rather than software development.

Key Characteristics:

- Usually **fixed-price**, but with **simpler terms** compared to development contracts.
- **End product:** Advisory reports or analysis-based recommendations.
- Typically **low financial risk**, so litigation or strict enforcement is rare.

Legal and Professional Considerations:

- **Confidentiality** is critical.
- The contract must clearly state **terms of reference**, defining:
 - Scope of work.
 - Boundaries of investigation.
- **Ambiguities** may arise if consultants wish to expand scope, but clients resist.

Quality Assurance:

- Enforcement relies on the **professional integrity and reputation** of the consultant.
- Not always possible to legally prove failure to meet expectations.



Legal & Ethical Considerations

- **Clarity & Transparency:** All terms must be clear and mutually understood
- **Data Privacy Compliance:** Must comply with laws like **GDPR, PECA**
- **Fair Terms:** Avoid exploitation of either party
- **IP Protection:** Ensure no code or product is copied illegally
- **Accountability:** Responsibilities must be clearly stated and followed



Lecture 5: Intellectual Property Rights in Computing

Definition

Intellectual property is any unique product of the human intellect that has commercial value.

In computing, this includes:

- Software
- Algorithms
- Source code
- User interfaces
- Documentation
- Multimedia content

Protecting Intellectual Property

Today, there are four different ways in which individuals and organizations protect their intellectual property: trade secrets, trademarks/service marks, patents, and copyrights.

Trade Secrets

Definition: A trade secret is a confidential piece of intellectual property that provides a company with a competitive advantage.

- Must be **actively protected** (e.g., NDAs, access controls)
- Example: **Coca-Cola syrup formula**
- Legal risks include **employee leaks** and **reverse engineering**, which are often legal

Trademarks and Service Marks

Definition:

A **trade mark** is any **graphically representable sign** capable of distinguishing one trader's goods or services from another's.

It may consist of:

- **Words** (including personal names)
- **Designs, letters, numerals**
- **Shapes** of goods or their **packaging**
Examples: Microsoft, IBM, Shell logo, Coca-Cola bottle.

Advantages of Registration

- Discourages rivals due to visibility in trade mark searches
- Stronger legal protection
- Nationwide coverage
- Monopoly can be **perpetual** (renewable)
- Can register after proving commercial success

Registration Criteria – Trade Marks Act 1994

Section 1: Definition

- Any sign that can be graphically represented and distinguish goods/services

Section 3: Absolute Grounds for Refusal

- Not graphically representable (e.g., smells)
- Lacks distinctiveness (unless acquired through use)

- Consists only of:
 - Common descriptive words (e.g., “Floppy Discs”)
 - Indications of type, quality, origin, value, etc.
- Becomes **customary** in trade language
- **Contrary to public policy**, deceptive, or illegal

Section 5: Relative Grounds for Refusal

- Same or similar to an existing mark
- Likelihood of confusion must be proven by opponent

Infringement of Registered Trade Marks – Section 10

Occurs when:

- A **confusingly similar** mark is used
- Public is misled about **association** between marks
- Use takes **unfair advantage** of or **damages** the mark’s character/reputation
Examples likely to trigger infringement: IBN, St Mitchell, Zerocks

Non-Infringing Acts

Use of a trade mark is **not infringement** if:

- Using one’s **own name or address**
- Describing product **kind, quality, purpose, origin**, etc.
- Referring to the mark for **spare parts**
 - Must follow **honest practice**, not take unfair advantage

Patents

Definition: A patent is a way the US government provides an inventor with an exclusive right to a piece of intellectual property.

- Protect inventions such as **new algorithms, encryption methods, and hardware mechanisms**
- Requires public disclosure; lasts **20 years**
- Grants **exclusive rights** to make, use, and sell the invention
- Cannot be renewed after expiration

What Can Be Patented

The Patents Act 1977 does not define “invention” but instead prescribes four mandatory criteria that must all be met before a patent may be granted (Section 1):

1. **Novelty**

- The invention must not have been disclosed or used publicly prior to the patent application date. Any prior public disclosure—even in obscure journals—destroys novelty. Confidential discussions (e.g. with bankers, lawyers, patent agents) should be governed by implied or, preferably, express obligations of confidence to avoid jeopardizing patent rights.

2. **Inventive Step**

- The claimed advance must not be obvious to a hypothetical skilled person in the field, who is assumed to be well-read and capable of literature searches but lacking in inventive originality. A convenient test is whether the solution would have been “very plain” to this hypothetical person without access to the patent specification.

3. **Industrial Application**

- The invention must have a practical or technical effect—that is, it must be capable of being made or used in any kind of industry.

4. **Excluded Subject Matter**

- The invention's subject matter must not fall within any class expressly excluded from patentability under Sections 1(2) and 1(3) (see below).

These requirements mirror Article 52 of the European Patent Convention and thus apply throughout most of Europe.

What Is Not Patentable

Sections 1(2) and 1(3) of the Patents Act carve out several categories of subject matter that may not be patented, either for public-policy reasons or because other forms of protection are more appropriate. Key exclusions include:

- **Scientific Theories**
 - Pure laws of nature or physics; however, a machine applying such a law may be patentable.
- **Mathematical Methods**
 - Abstract calculation techniques (e.g. algorithms for square roots), unless embodied in a practical apparatus.
- **Aesthetic Creations**
 - Literary, dramatic, musical or artistic works; these are protected by copyright when fixed in material form.
- **Presentation of Information**
 - Data layouts or tables, which fall under copyright or database rights.
- **Schemes, Rules or Methods**
 - Methods for mental acts, games, business operations, or “programs for a computer” as such.



Copyrights

Definition: A copyright is how the US government provides authors with certain rights to original works that they have written.

- Applies to **original written or digital works**, including:
 1. Source code
 2. Documentation
 3. UI/UX layouts
 4. Screen displays
- Gives five major rights:
 1. Reproduce
 2. Distribute
 3. Display publicly
 4. Perform publicly
 5. Create derivative works

Fair Use Doctrine

Definition: The right given to a copyright owner to reproduce a work is a limited right. Under some circumstances, called fair use, it is legal to reproduce a copyrighted work without the permission of the copyright holder.

Allows limited use of copyrighted material without permission for:

- Teaching and research

- Criticism and commentary
- News reporting

Fair Use Factors:

1. Purpose (educational vs. commercial)
2. Nature (non-fiction preferred)
3. Amount (short excerpts allowed)
4. Effect on the market (less market impact = more permissible)

Peer-to-Peer Networks and Cyberlockers

On the Internet, the adjective peer-to-peer refers to a transient network allowing computers running the same networking program to connect with each other and access files stored on each other's hard drives. Peer-to-peer networks stimulate the exchange of data in three ways:

1. First, they give each user access to data stored on many other computers.
2. Second, they support simultaneous file transfers among arbitrary pairs of computers.
3. Third, they allow users to identify those systems that will be able to deliver the desired data more rapidly, perhaps because they have a faster Internet connection or are fewer routing hops away.

Cyberlockers (also called file-hosting services or cloud storage services) are Internet-based file-sharing services that allow users to upload password-protected files. Users can give other people access to the files they have uploaded by sharing passwords. People who wish to collaborate on a project often find sharing large files through cyberlockers more convenient than sending them back and forth as attachments to email messages. However, cyberlockers also make it easy for people to share copyrighted material, such as songs and movies. In addition, cyberlocker use is much more difficult for government officials to track than peer-to-peer file sharing.

Protections for Software

In the early days of the computer industry, there was no strong demand for intellectual property protection for software. Most commercial software was produced by the same companies



manufacturing computer hardware. They sold complete systems to customers, and the licensing agreements covered use of the software as well as the hardware.

Software Copyrights

The first software copyrights were applied for in 1964. The Copyright Office allowed the submitted computer programs to be registered, reasoning that a computer program is like a “how-to” book. When a piece of software gets copyright protection, what exactly is copyrighted?

1. First, copyright protects the expression of an idea, not the idea itself. For example, suppose you develop a program for a relational database management system. You may be able to copyright your implementation of a relational database management system, but you cannot copyright the concept of using relational databases to store information.
2. Second, copyright usually protects the object (executable) program, not the source program. Typically, the source code to a program is confidential; in other words, a trade secret of the enterprise that developed it. The company only distributes the object program to its customers. The copyright also protects the screen displays produced by the program as it executes. This is particularly valuable for the developers of video games.

However, doing any of the following actions without authorization of the copyright holder is a violation of copyright law:

1. Copying a program on to a CD to give or sell to someone else.
2. Preloading a program onto the hard disk of a computer being sold.
3. Distributing a program over the Internet.

Open-Source Software

Open source is an alternative way of distributing software. Licenses for open-source programs have the following key characteristics:

1. There are no restrictions preventing others from selling or giving away the software.
2. The source code to the program must be included in the distribution or easily available by other means (such as downloadable from the Internet).
3. There are no restrictions preventing people from modifying the source code, and derived works can be distributed according to the same license terms as the original program.
4. There are no restrictions regarding how people can use the software.
5. These rights apply to everyone receiving redistributions of the software without the need for additional licensing agreements.

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6. The license cannot put restrictions on other software that is part of the same distribution. For example, a program's open-source license cannot require all of the other programs on the CD to be open source.

Legal Protection of Confidential Information in Software Engineering

Confidential information—from proprietary code and client data to strategic know-how—is a cornerstone asset for software houses. This topic examines the legal mechanisms that safeguard such information, balancing a company's need to protect trade secrets with employees' mobility and the public interest.

1. Definition and Scope of Confidential Information

- **What Counts as “Confidential”?**
 - Lord Greene's formulation: information “not public property or knowledge”
 - Encompasses programs, algorithms, data structures, business processes, client specifications, etc.
- **Why It Matters in Software**
 - Prevents leakage of competitive advantage
 - Preserves patentability (pre-filing secrecy)

2. Three-Part Test for Breach of Confidence

1. **Secrecy:** Information must have sufficient quality of confidence.
2. **Circumstances of Disclosure:** Must be imparted in a context imposing an obligation of confidence.
3. **Unauthorized Use/Disclosure:** Actual or threatened misuse of the information.

3. Sources of the Obligation of Confidence



3.1 Express Contractual Clauses

- **Standalone Non-Disclosure Agreements (NDAs)** vs. confidentiality clauses baked into employment, consultancy, and service contracts
- **Key Terms:** scope of use, duration, carve-outs (public domain, prior knowledge, third-party source)
- **“Black-Box” Contracts:** physical inspection restrictions (e.g. Paul v. Southern Instruments)

3.2 Implied Contractual Obligations

- Implied duty of good faith and fidelity in employment contracts
- Supplementing express clauses (Thomas Marshall v. Guinle)

3.3 Equitable Duty of Confidence

- Even absent any contract
- Arises when information is shared in confidence for a limited purpose

4. Special Contexts for Confidentiality

4.1 Employees vs. Independent Contractors

- **Post-Termination Duties**
 - Ex-employees remain bound to protect trade secrets—cannot use information beyond their “general skill and knowledge” (Faccenda Chicken v. Fowler)
 - Independent contractors may face a stricter regime, particularly in long-term engagements
- **Restrictive Covenants & “Garden Leave”**

- Reasonable time- and scope-limited restraints on post-employment competition
- Prolonged notice plus paid garden leave to neutralize knowledge decay

4.2 Third-Party Recipients

- Equitable obligation extends to those who knowingly receive wrongly disclosed secrets
- Tortious liability for inducing breach of confidence

Copyright Protection for Software and Digital Works

Copyright is the foundational legal mechanism safeguarding the creative form in which ideas—literary text, code, diagrams, databases, multimedia—are expressed. In the digital age, copyright faces both unprecedented opportunities (global distribution, new markets) and threats (mass piracy, uncontrolled online copying). This topic examines the scope, mechanics, and challenges of copyright as it applies to software and related digital works.

1. Fundamentals of Copyright

- **Definition of “Work”** (CDPA 1988):
 1. **Literary works:** novels, manuals, computer programs, compilations, databases
 2. **Dramatic, musical, artistic works:** screenplays, designs, graphics, multimedia content
 3. **Sound recordings, films, broadcasts, cable programmes**
 4. **Typographical arrangements** of published editions
- **Automatic Protection:** No registration required—copyright subsists on creation (fixation in any material form).

2. Six Exclusive Rights of Copyright Owners

- 
1. **Reproduction:** copying or storage by any means (including transient electronic copies)
 2. **Issuing Copies:** distribution to the public
 3. **Rental & Lending:** commercial hire or library-style lending
 4. **Public Performance/Display**
 5. **Broadcasting & Cable Transmission**
 6. **Adaptation:** translations, compilations, or any derivative work

3. Special Protections for Software & Databases

- **Computer Programs** (1992 Regulations): treated as literary works; preparatory design material also protected
- **Databases** (1997 Regulations):
 - **Copyright** in original selection or arrangement (“author’s own intellectual creation”)
 - **Sui Generis Database Right:** 15-year protection for substantial investment in obtaining, verifying or presenting contents

4. Authorship & First Ownership

- **General Rule:** author = creator of the work
- **Special Cases:**
 - **Computer-generated works:** “maker”—the person who arranges for its creation
 - **Employees:** employer owns copyright in works made “in the course of employment” (s.11 CDPA)

- **Independent Contractors:** retain copyright unless assignment agreed

5. Duration of Protection

Category	Term
Literary, dramatic, musical, artistic	Life of author + 70 years
Computer-generated works	50 years from year of creation
Sound recordings	50 years from year of recording or release
Films	70 years from death of last surviving key creator (director, screenwriter, composer)
Broadcasts & cable programmes	50 years from broadcast date
Typographical arrangements	25 years from first publication

- **Directive Influence:** EU Duration Directive harmonized and extended terms

6. Scope & Limits: Exceptions and Fair Dealing

- **Statutory Exceptions:**
 - Private copying, quotation, criticism, reporting
 - Backup copies of software; interoperability reverse-engineering
- **Digital Challenges:**

- Online streaming vs. copying
- Technological protection measures (DRM) and “anti-circumvention” rules

Infringement of Copyright: Primary and Secondary Liability

Copyright owners enjoy six exclusive rights—reproduction, distribution, rental/lending, public performance/display, broadcasting/cable transmission, and adaptation. Infringement occurs when others exercise these rights without permission. This topic examines the two categories of infringement under UK law—primary and secondary—including the core concepts, key tests, and modern challenges in digital environments.

1. Primary Infringement

1.1 Definition

Any unauthorized exercise of one of the six exclusive rights named in the Copyright, Designs and Patents Act 1988 (CDPA):

1. **Copy** (reproduce in any material form, including transient electronic copies)
2. **Issue** copies to the public (distribution right)
3. **Rent or lend** copies to the public
4. **Perform/play/show** the work in public
5. **Broadcast or include** in a cable programme service
6. **Adapt** the work or exercise any of the above on an adaptation

Note: Knowledge of copyright status is irrelevant; innocent infringement is still primary infringement (civil only).

1.2 Non-Literal Copying & Idea/Expression Dichotomy

- **Non-literal copying:** reproducing program structure, sequence, organization (SSO) without verbatim code.

- **Idea vs. Expression:**

- **Directive (93/98/EEC):** only expression is protected; underlying ideas/principles are free.

2. Secondary Infringement

2.1 Definition & Criminal Exposure

Secondary infringement involves dealing with infringing copies or assisting others to infringe, **with knowledge** or reason to believe infringement has occurred. It can attract **criminal sanctions** (fines, imprisonment) as well as civil remedies.

2.2 Key Offenses

- **Importing, selling, or renting** infringing copies in the course of business
- **Possession** of infringing copies for commercial advantage
- **Supplying** devices or services (e.g., peer-to-peer networks) designed to enable infringement
- **Facilitating public performance** (providing premises or apparatus knowing infringement will occur)

Moral Rights

Definition:

Moral rights are personal rights granted to authors under the **Copyright, Designs and Patents Act 1988**. They are distinct from economic copyright rights but are protected under similar legal principles.

Key Moral Rights:

1. Paternity Right

- The right of an author to be **identified as the author** of their work.

- Traditionally called the “paternity right” (now considered a gender-biased term).

2. Integrity Right

- The right to **object to derogatory treatment** of the work that is prejudicial to the author’s **honour or reputation**.

3. Right Against False Attribution

- The right of any person to **object if falsely named** as the author of a **literary, dramatic, musical, artistic work**, or a **computer program**.

Limitations for Software Engineers:

- **Paternity and Integrity rights do not apply to:**
 - Authors of **computer programs**
 - **Computer-generated works**
- These rights **can apply to:**
 - Authors of **related works** such as a **printed computer manual**

Domain Names and Cyber-Squatting

Domain Names and Trade Marks: The Connection

- Consumers **expect domain names** (e.g., **marksandspencer .com**, **marksandspencer .co .uk**) to reflect **brand names** or **registered trade marks**.
- Mismatch or misuse of these domain names may cause **consumer confusion**, reputational damage, or even financial loss.

The Problem of Cybersquatting

- **Cybersquatting:** Registering well-known domain names without intending to use them—solely to sell them to legitimate businesses later.

- Example: In **Harrods Ltd v UK Network Services Ltd (1996)**, defendants registered **harrods.com** intending to sell it to Harrods.
- Similar actions were taken against cybersquatters by **Marks & Spencer** and **Burger King**.

Ethical Considerations in IPR

- Respect creator rights and avoid piracy
- Understand and comply with software licenses
- Acknowledge sources (no plagiarism)
- Ensure clients do not unintentionally violate IPR laws
- Promote innovation through **responsible use** and **fair distribution**

Lecture 6: Employee Relations Law in the IT Industry

Topics Covered: The Framework of Employee Relations Law and Changing Management Practices

Introduction to Employee Relations (ER) Law

Employee relations refers to the legal, ethical, and professional framework that governs the **relationship between an organization and its employees**. It encompasses communication, workplace rights, employee engagement, performance management, dispute resolution, and organizational policies.

A strong employee relationship system fosters:

- **Trust and respect**
- **Legal compliance**
- **Increased productivity**
- **Reduced conflict and turnover**

Basis of Employee Relationship

Basis	Description
Pay (Compensation)	Fair, timely, and performance-based wages and benefits
Work Bargain	The mutual exchange : employees provide labor; employers provide compensation, security, and working conditions
Voice & Participation	Systems that allow employees to express opinions , offer feedback, and be involved in decisions

Employment Practices	Recruitment, training, promotion, discipline, and termination processes must be fair, documented, and transparent
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Types of ER

Relationship Type	Description
Employee to Employer	Based on contracts, duties, responsibilities, and rights
Employee to Manager	Performance, guidance, feedback, and leadership relationship
Employee to Employee	Peer collaboration, team dynamics, mutual respect
Employee to HR Department	Handling of grievances, benefits, training, and career development
Employee to Union/Representation	In collective settings, for bargaining, dispute handling

Factors Influencing ER

Factor	Explanation
Organizational Culture	The values, communication style, and leadership approach shape relationships
Management Style	Autocratic vs. participative styles affect trust and engagement
Work Environment	Physical and psychological conditions (safe, inclusive, supportive)

Job Satisfaction	Career growth, recognition, and compensation levels influence morale
Communication	Open, two-way communication builds trust and reduces conflict
Technology	Digital tools, surveillance, and remote work affect autonomy and trust
Economic Conditions	Inflation, layoffs, or market instability can cause strain or disengagement

Employee Rights

Right	Explanation
Right to Fair Pay	Minimum wage, equal pay for equal work, and overtime pay
Right to Privacy	Protection of personal data and freedom from unjust surveillance
Right to Safety	Secure and hazard-free work environment
Right to Equality	Protection against discrimination (gender, religion, ethnicity, etc.)
Right to Unionize	Join trade unions, collective bargaining, strike (subject to legal limits)
Right to Due Process	Fair treatment in promotion, discipline, and termination
Right to Speak Out	Whistleblower protection and employee voice platforms



Key Elements of Employee Relations Legal Framework

♦ Employment Contracts

Contracts are fundamental in employment law. As outlined in your reference slides, a contract:

- Defines mutual obligations (duties, benefits, working hours)
- Is enforceable under law
- Should be **clear, consistent, and unambiguous** to prevent conflict

A good employment contract ensures:

- Job role clarity
- Compensation and benefits
- Termination clauses
- Confidentiality agreements
- Intellectual property ownership (where applicable)

♦ Labor Laws

National labor codes typically include:

- **Minimum wage laws**
- **Work hour limits**
- **Health & safety requirements**
- **Unionization rights**

- **Leave entitlements (sick, maternity, annual)**

Example: In Pakistan, laws like the Industrial Relations Act (2012) govern labor union activity, while the Factories Act (1934) regulates working conditions.

♦ Dispute Resolution

Contracts and law must include **mechanisms for arbitration**, grievance redressal, and procedures for **constructive dismissal**, ensuring due process before any disciplinary action is taken.

Equal Pay and Sex Discrimination

A. Introduction

Equal pay and sex discrimination are critical aspects of modern employee relations and employment law, particularly as they pertain to gender equality in the workplace. Although progress has been made—especially due to European Union (EU) directives—the wage gap between men and women persists, and underlying systemic issues remain a significant barrier to true equality.

B. Legal Background and Framework

1. European Union Influence

- The **UK amended its equal pay laws** to conform with European Community legislation, specifically adopting the principle of “**equal pay for work of equal value.**”
- Enforcement mechanisms were also mandated, ensuring a legal route for women to challenge wage inequality.

2. Social Charter Directives

- The **Social Charter of the EU** introduced directives focused on:
 - Atypical employment (e.g., part-time, temporary roles).

- Indirect discrimination where **part-time workers (majority female)** received less favourable treatment than full-time counterparts.
- **European Court of Justice (ECJ)** has upheld equal treatment claims in areas such as:
 - Occupational pensions
 - Sickness and severance benefits
 - Pay adjustments

C. Persistence of the Gender Pay Gap

1. Current Statistics and Trends

- The UK continues to have **one of the widest gender wage gaps in the EU**, despite legislative advances.
- Economists debate whether this gap stems primarily from **discrimination or occupational segregation**.

2. Occupational Segregation

- **Segregation** refers to women being concentrated in **lower-paid, part-time, or temporary jobs**, especially in sectors like retail and hospitality.

3. Rise of Part-Time Employment

- Post-WWII employment trends show the **substitution of full-time jobs with part-time roles for women**.
- Research (Robinson & Wallace) indicates **no direct wage discrimination** against part-time women workers in base pay—but highlights persistent **segregation and limited upward mobility**.

D. Enforcement Challenges

- The process of **filing an equal pay claim** is legally **complex, time-consuming, and expensive**.

- This discourages **low-paid or part-time female workers** from pursuing justice under existing frameworks.
- Enforcement mechanisms need to become more accessible and practical to achieve real equality.

E. Technology Sector and Gender Disparities

1. Lack of Female Representation in IT

- There is a **consistently low intake of women** in IT and computer science fields in the UK.
- Key barriers:
 - The **male-dominated image** of IT.
 - The **negative public perception** due to job automation and job losses.
 - Lack of early **positive engagement with IT** in education and public life.

2. Opportunities in IT for Women

Despite underrepresentation, the **IT sector offers unique advantages** for promoting gender equality:

- **Flexible working arrangements** (e.g., remote work, part-time schedules, project-based roles).
- **Support for career breaks** and **non-linear career paths**.
- Digital tools and infrastructure that support **remote and asynchronous collaboration**.

3. Future Directions

- The IT industry needs a **rebranding effort** to highlight its **societal value, creativity, and flexibility**, especially to appeal to young women.

- Encouraging female participation in tech not only helps address pay disparity but also **diversifies the talent pool** and fosters **innovation through inclusion**.

Changing Management Practices in Employee Relations

As the workplace evolves, so do management strategies. Traditional command-and-control models are giving way to **employee-centric**, **tech-enabled**, and **values-driven** approaches.

♦ Performance Management Evolution

- From annual reviews → continuous feedback
- KPIs, 360-degree feedback, real-time dashboards

♦ Digital Transformation in HR

- Use of AI in recruitment, HR chatbots, analytics
- Raises concerns of **bias**, **privacy**, and **automation ethics**

♦ Flexible Work Structures

- Rise of **remote work**, **freelancing**, **gig roles**
- Requires flexible contracts, digital safety protocols, and wellness support

♦ Inclusive Leadership & Diversity Management

- DEI (Diversity, Equity, Inclusion) training and hiring policies
- Safe spaces for underrepresented groups
- Anti-harassment systems



♦ Employee Wellbeing & Mental Health

- Recognition of emotional burnout, stress
- Inclusion of mental health resources, support systems

♦ Employee Voice & Engagement

- Anonymous surveys, suggestion platforms, open-door policies
- Participative decision-making in strategic directions

Ethical Considerations in Management Practices

Drawing from your material on **professional and contractual ethics**:

- Managers must operate with:
 - **Integrity:** Full disclosure of role expectations and benefits
 - **Fairness:** Equitable performance evaluation and promotion
 - **Respect:** For cultural, gender, and professional diversity
 - **Accountability:** Owning the outcomes of leadership decisions

Lecture 7: Human Resource Management in IT

Topics Covered: Human Resource Management in IT

Introduction

Human Resource Management (HRM) in the IT sector involves the strategic planning and execution of functions such as **recruitment, training, performance evaluation, employee engagement, and ethics compliance**, specifically tailored to the unique needs of **technology-driven organizations**.

Computer professionals have special responsibilities not only to their clients but to the general public... thus, HR must ensure ethical behavior and competence in technology domains.

Strategic HR Functions in the IT Sector

Function	Role in IT
Recruitment	Identifying talent with up-to-date programming, networking, AI/ML, and cybersecurity skills
Training	Continuous learning programs for evolving tech (cloud, DevOps, agile, blockchain)
Performance Management	Using KPIs, agile metrics, and digital dashboards for feedback and appraisals
Compensation	Salary benchmarking based on global trends; includes stock options and remote incentives
Retention & Engagement	Focusing on culture, innovation, flexibility, and career progression
Ethical Oversight	Enforcing data privacy, AI transparency, digital behavior norms

Key Challenges in HRM for IT Organizations

♦ Talent Scarcity

- Difficulty in hiring skilled professionals for fast-evolving tech domains.
- Ethical hiring requires transparency and anti-bias in AI-driven recruitment tools.

♦ Attrition and Remote Work

- The global tech industry sees high turnover. HR must support:
 - Flexible hours
 - Mental health initiatives
 - Digital team building

♦ Ethics and Conduct

- HR must implement:
 - Cybersecurity behavior policies
 - Non-disclosure and IP protection
 - Guidelines for AI fairness and algorithmic bias

The HRM function must ensure employees act ethically while handling confidential data, algorithms, and customer information.

Technology's Impact on HR Management

Aspect	Digital Transformation
Recruitment	AI resume screening, video interviews, predictive hiring

Learning & Development	LMS, MOOCs, AR/VR simulations, GitHub integration
HR Analytics	Big data used to analyze productivity, engagement, retention
Onboarding & Compliance	E-contracts, automated policy training, LMS certifications
Monitoring Tools	Ethical surveillance of productivity via keystroke/activity trackers (with informed consent)

Ethical Risk: Use of AI in hiring, monitoring, and evaluation must be **transparent, auditable**, and **free from algorithmic discrimination**

Ethical and Legal Responsibilities of HR in IT

HR must ensure:

- Protection of **employee and customer data**
- Clear **ownership rights** of software and creative output (IPR)
- Zero tolerance toward:
 - Discrimination in tech teams
 - Digital harassment
 - Uninformed data collection

Software copyrights, licensing rules, and contract clarity are essential for HR in software firms to manage professional liability.

Lecture 8: Health and Safety at Work – In the IT Industry

Topics Covered: Health and Safety at Work

Introduction

Health and Safety at Work refers to the **legal, ethical, and organizational responsibility** to provide a safe, healthy, and hazard-free working environment. It is a key component of **employee welfare, risk management, and legal compliance** across all industries, including the IT sector.

Professionals must take reasonable steps to ensure the health and safety of people affected by their work.

Legal and Ethical Basis

Most countries enforce **Occupational Health and Safety (OHS)** laws that:

- Mandate safe physical working environments
- Require reporting of accidents/injuries
- Enforce penalties for non-compliance
- Protect whistleblowers who report unsafe conditions

Legal Examples:

Country	Law Example
Pakistan	Factories Act, 1934 / Shops & Establishments Ordinance
UK	Health and Safety at Work Act, 1974
USA	Occupational Safety and Health Act (OSHA), 1970



Employer Responsibilities

Employers must ensure:

- Risk assessments are carried out regularly
- Proper safety equipment and training are provided
- Emergency procedures are in place (fire drills, exits, first aid)
- Employees are not exposed to hazardous conditions (physically or digitally)
- Employees are informed about health and safety policies

The ethical responsibility goes beyond compliance; employers must foresee risks even before laws demand it.

Employee Responsibilities

Employees are responsible for:

- Following safety instructions and procedures
- Using equipment responsibly
- Reporting unsafe practices or incidents
- Wearing protective gear where applicable
- Participating in safety training sessions

Health & Safety in the IT Workplace

Although the IT environment is often considered low-risk compared to manufacturing, **health and safety risks still exist**, especially related to **ergonomics, mental health, and digital burnout**.

Common IT-Specific Risks:

Risk	Description
Repetitive Strain Injury (RSI)	Due to long keyboard/mouse usage
Back and Neck Pain	Poor posture, non-ergonomic chairs
Eye Strain / Headaches	Extended screen time without breaks
Mental Health Issues	Burnout, overwork, digital overload
Sedentary Lifestyle Risks	Obesity, cardiovascular problems

Risk Mitigation Measures

Ergonomics:

- Use of adjustable chairs and desks
- Proper monitor height and distance
- Wrist rests and foot support

Digital Health:

- Enforcing screen breaks (20-20-20 rule)
- Encouraging time off and disconnecting after work hours
- Promoting digital minimalism and work-life boundaries



Workplace Environment:

- Adequate lighting and ventilation
- Safe cabling and electrical systems
- Quiet zones for focused work

Emergency Preparedness:

- Fire extinguishers, alarms, evacuation maps
- First-aid kits and trained responders
- Cybersecurity protocols (for digital infrastructure safety)

Ethical Perspective on Health and Safety

According to **Baase** and **Kizza**:

- Employers must prioritize human well-being over profits
- Ignoring long-term health risks (e.g., RSI, mental burnout) is unethical
- Transparency about hazards and mitigation is vital
- Workers should not be punished for refusing unsafe work

A professional's obligation to public health and safety must override client or employer demands if harm is foreseeable.

Lecture 9: Software Liability & Accountability

Topics Covered: Software Liability, Liability and Practice

What is Software Liability?

Software liability refers to the legal and ethical responsibility of software developers, engineers, or companies for **harms caused by defects, failures, or misuse of software systems**.

As software increasingly controls life-critical systems (healthcare, finance, aviation), legal expectations for **professional care and reliability** are rising.

If a computer professional creates a defective system that causes injury, the question is not just who is to blame, but who is legally liable.

Legal Foundations of Liability

There are **three major bases** for assigning liability in software and systems engineering:

A. Strict Liability

- Holds producers liable for harm caused by **defective products**, even if they were not negligent.
- Example: A software glitch in an autonomous car causes an accident.

B. Negligence

- Arises when harm is caused by a **failure to meet a reasonable standard of care**.
- Example: A developer fails to test security code, resulting in a data breach.

C. Breach of Warranty / Contract

- If a product fails to perform as promised under a license or contract, the customer can sue for damages.

- Example: A software service advertised as "99.99% uptime" experiences repeated outages.

Software developers do not enjoy the same immunity that manufacturers of physical goods once had. Increasingly, courts view software as a product.

Challenges in Applying Liability to Software

♦ Software Is Intangible

- Unlike physical products, software evolves through updates, patches, and configurations.

♦ Complexity and Bugs Are Inevitable

- No complex software is ever "perfectly bug-free."
- But **failure to test adequately** may still lead to liability.

♦ User Misuse or Modification

- Liability may be reduced if users alter or misuse the software in unintended ways.

♦ Third-Party Components

- Use of open-source or third-party libraries complicates the chain of responsibility.

Practices That Reduce Software Liability

Best Practice	Impact
Thorough Testing and QA	Reduces risk of undetected bugs in critical systems
Clear Licensing Agreements	Defines usage rights, disclaimers, and responsibilities
Professional Certification & Standards	Demonstrates adherence to recognized best practices (e.g., ISO, IEEE)
User Training and Documentation	Helps avoid misuse and informs users of limitations

Logging and Auditing	Provides evidence trail in case of failure or legal investigation
Ethical Risk Assessment	Identifies how the system could cause harm (safety, privacy, security)

Liability cannot be avoided simply by disclaimers. Professionals have a duty to anticipate risks and warn clients.

Professional Liability vs. Product Liability

Type	Explanation
Product Liability	Focuses on the software product being defective or dangerous
Professional Liability	Focuses on developer actions or negligence during software creation

Professional engineers and software developers may be held to the same standards as **medical doctors or civil engineers**, especially in critical domains.

Software professionals should not release systems unless they are convinced the systems are safe.

Lecture 10: Cybercrime, Computer Misuse & Legal Control

Topics Covered: Computer Misuse and the Criminal Law, Regulation and Control of Personal Information.

Computer Misuse

Computer Misuse refers to **unauthorized, unethical, or illegal use of computers, systems, or networks** that causes harm, loss, or disruption to individuals, organizations, or society. Misuse includes any act that breaches the privacy, integrity, or availability of a computer system. It is often addressed in laws like the **Computer Misuse Act 1990 (UK)** and **cybercrime laws globally**, and covers a wide spectrum, from unauthorized access to data theft and cyberterrorism.

Types of Misuse

Type of Misuse	Description
Unauthorized Access (Hacking)	Accessing computer systems or networks without permission
Malware Distribution	Creating or spreading viruses, worms, ransomware, spyware, etc.
Data Theft	Illegally copying, transmitting, or using private or confidential data
Denial-of-Service (DoS) Attacks	Overloading a server to make services unavailable
Phishing / Social Engineering	Deceiving users to gain sensitive information like passwords
Intellectual Property Infringement	Copying or distributing software, music, videos without permission

Cyberbullying / Online Harassment	Using digital means to threaten, abuse, or harass individuals
System Sabotage / Insider Threats	Damaging or destroying IT systems intentionally, especially by employees
Digital Fraud / Identity Theft	Using someone else's digital identity for financial or personal gain

Ethical Perspective

From an ethical standpoint, misuse of computer systems:

- Violates **privacy** and **trust**
- Causes **economic and emotional damage**
- Threatens **national security and public safety**
- Undermines **confidence** in technology

Just because something is technologically possible does not make it ethically permissible.

Causes of Computer Misuse

- Lack of cyber awareness
- Weak security practices in organizations
- Insider threats and disgruntled employees
- Greed, revenge, or ideological motives
- Poor law enforcement and outdated laws
- Easy access to hacking tools and dark web resources

Responsibility of Computing Professionals

According to the **ACM and IEEE Codes of Ethics**:

- Professionals must **prevent the misuse** of systems they develop or maintain
- Must report vulnerabilities and unethical use
- Must protect users' rights and digital privacy
- Must avoid developing tools meant for harm (e.g., spyware, hacking utilities)

An engineer has a moral duty not to knowingly develop systems that can be easily misused or cause harm.

Preventing and Mitigating Misuse

Organizational Practices:

- Strong access controls and authentication mechanisms
- Employee training in cybersecurity and data ethics
- Regular audits and penetration testing
- Clear policies on acceptable use of IT resources

Technical Measures:

- Firewalls, antivirus, and intrusion detection systems (IDS)
- Data encryption and secure storage
- Backup and disaster recovery plans
- Role-based access and logging

Legal and Regulatory:

- Strict enforcement of cyber laws
- International cooperation for cybercrime investigation
- Reporting mechanisms for victims

Criminal Law in Computing

What is Criminal Law?

Criminal law refers to the **body of laws that define criminal offenses**, prescribe penalties, and regulate the procedures for investigation, prosecution, and punishment of crimes. In computing, **criminal law governs acts involving unauthorized access, fraud, data theft, and cybercrimes**.

Laws must evolve as technology evolves. Criminal law is increasingly central in regulating digital behavior.

Relevance in Computing

With the expansion of the digital world, criminal law now covers a wide range of **computer-related offenses**, often referred to as **cybercrime**.

Common Computer-Related Crimes:

Type of Crime	Description
Hacking	Unauthorized access to computer systems
Malware Distribution	Creation or spread of viruses, worms, or ransomware
Identity Theft	Stealing personal data to impersonate others

Phishing & Social Engineering	Fraudulently obtaining information
Denial-of-Service (DoS) Attacks	Disrupting access to a service
Software Piracy	Unauthorized copying and distribution of software
Cyberstalking / Harassment	Using technology to stalk, harass, or threaten
Data Breaches	Exposure of personal or corporate data
Online Fraud / Financial Theft	Use of online systems to steal money or credentials

Elements of a Criminal Offense (in IT Context)

For an act to qualify as a **criminal offense**, it typically must meet two elements:

Legal Element	Meaning in IT Context
Actus Reus (guilty act)	A deliberate, illegal act such as unauthorized access or theft of data
Mens Rea (guilty mind)	The intention to commit the crime (e.g., knowingly bypassing authentication)

Key Cybercrime Laws Globally

Country	Law Example
Pakistan	<i>Prevention of Electronic Crimes Act (PECA), 2016</i>

USA	<i>Computer Fraud and Abuse Act (CFAA)</i>
UK	<i>Computer Misuse Act, 1990</i>
India	<i>Information Technology Act, 2000</i>
EU	<i>General Data Protection Regulation (GDPR)</i> (criminal liability in data breaches)

Investigation and Enforcement

Criminal cases involve:

- **Digital forensics:** Tracing logs, IP addresses, device footprints
- **Jurisdictional complexity:** Offenders often operate across borders
- **Chain of custody:** Legal process to preserve evidence integrity
- **Cybercrime cells:** Specialized law enforcement units

Prosecution is often hindered by jurisdictional challenges and a lack of digital evidence handling expertise.

Ethical and Legal Balance

Ethical Dilemmas in Criminal Law:

- **Is ethical hacking a crime?**
Intent matters. Ethical hacking with permission (e.g., penetration testing) is legal.
- **What about software tools used for both good and bad purposes (dual-use)?**
Tools themselves may not be illegal, but their use can be.
- **Freedom of expression vs hate speech?**
Social media posts can fall under criminal law if they incite violence or hate.

Not all unethical actions are illegal, and not all illegal actions are unethical. But professionals must understand both dimensions.

Regulation and Control of Personal Information

Definition

Personal information refers to any data that can identify an individual, either directly or indirectly, such as names, addresses, biometric data, browsing history, financial records, health status, etc.

The regulation and control of personal information refers to legal, ethical, and technical measures aimed at protecting individuals' privacy from unauthorized access, use, or distribution.

Why Regulation is Needed

- **Mass data collection** via smartphones, social media, IoT, and surveillance cameras
- Rise in **identity theft**, **data breaches**, and **cybercrime**
- Individuals often lack **awareness** or **control** over how their data is used
- Companies and governments may **exploit personal data** for profit or surveillance

Technology enables collection and dissemination of personal data in ways never possible before, necessitating ethical and legal safeguards.

Technical and Organizational Control Measures

Method	Description
Encryption	Secures data in storage and transit
Access Control	Limits who can view or modify data
Anonymization	Removes personally identifiable information

Audit Trails & Logs	Track access and changes
User Consent Mechanisms	Informed and explicit opt-in/opt-out
Data Minimization	Collect only what is necessary

Challenges in Controlling Personal Information

- **Big Tech monopolies** with massive user data
- **Cross-border data flow** creates jurisdiction issues
- **Cloud computing** makes it harder to localize storage
- **Lack of awareness** among users about privacy settings
- **Dark patterns** used to trick users into giving consent

Ethical Principles from Reference Texts

According to *Quinn*:

- **Respect for autonomy**: Individuals must control their own data
- **Justice and fairness**: Data should not be used to discriminate
- **Non-maleficence**: Organizations must prevent harm (e.g., breaches)

According to *Baase*:

- **Informed consent** is key
- **Transparency** is essential for ethical use
- Even anonymized data can sometimes be re-identified

According to *Kizza*:

- Calls for **international coordination** of data regulation
- Emphasizes **cyber ethics education** as a preventive measure

Major Laws and Regulatory Frameworks

General Data Protection Regulation (GDPR) – EU

- Introduced in 2018
- Gives individuals greater control over their data
- Applies to all organizations dealing with EU residents' data

Key Rights:

- Right to access, correct, and delete data
- Right to data portability
- Right to be informed
- Right to object to processing
- Right to compensation for misuse

California Consumer Privacy Act (CCPA) – USA

- Similar to GDPR, applies to businesses in California
- Focuses on data collection transparency and opt-out rights

Pakistan's Draft Data Protection Bill

- Seeks to regulate processing of personal data
- Establishes a Data Protection Authority

- 
- Mandates breach notification and consent-based data collection

Lecture 11: Global Codes of Ethics in IT

Topics Covered: Overview of the British Computer Society Code of Conduct, IEEE Code of Ethics, ACM Code of Ethics and Professional Conduct, ACM/IEEE Software Engineering Code of Ethics and Professional Practice.

British Computer Society (BCS) Code of Conduct

Introduction

The **British Computer Society (BCS)**—also known as *The Chartered Institute for IT*—is the UK's leading professional body for computing. It has established a **Code of Conduct** that sets ethical and professional standards for its members.

The BCS Code of Conduct defines the professional and ethical standards required of all members and registrants in the field of computing.

Purpose of the BCS Code

The Code:

- Ensures **public confidence** in IT professionals
- Promotes **ethical behavior** in professional practice
- Provides a **framework for accountability**
- Protects **the integrity and reputation** of the IT profession

Core Principles of the BCS Code of Conduct

The BCS Code is structured around **four key principles** that members must adhere to:

1. Public Interest

Members shall have due regard for public health, privacy, security and wellbeing.

Members must:

- Prioritize the **public good** over personal or organizational gain
- Protect **privacy** and **confidential information**
- Promote **equal access** to IT and avoid discriminatory practices
- Report unethical or illegal conduct that could harm the public

2. Professional Competence and Integrity

Members shall act with integrity and maintain professional competence.

Members must:

- Perform services **only within their area of competence**
- Engage in **continuous professional development (CPD)**
- **Avoid conflicts of interest** and declare them when they arise
- Be **honest** about qualifications, experience, and capabilities

3. Duty to Relevant Authority

Members shall uphold the reputation and good standing of the profession and the relevant authority.

Members must:

- Comply with relevant **laws, regulations, and policies**
- **Respect their employer's or client's interests**, unless they conflict with the public good
- Avoid bringing the **profession or organization into disrepute**
- Cooperate with **investigations** into ethical violations or complaints

4. Duty to the Profession

Members shall uphold the reputation of the profession and shall seek to improve standards of practice.

Members must:

- Promote and encourage **high standards** in the profession
- **Mentor and support** other professionals
- Share knowledge to **advance the field**
- Take action when observing **unethical behavior** by peers

Disciplinary Procedures

- The BCS has a **Professional Conduct Panel** to investigate breaches
- Sanctions can include **warnings, suspensions, or revocation of membership**
- Members have the right to a **fair hearing and appeal**

Importance of the BCS Code in Computing Ethics

The Code:

- Establishes **trust between IT professionals and society**
- Promotes **responsible development and use of technology**
- Provides a **basis for resolving ethical dilemmas**
- Encourages **professionalism and accountability** in computing roles



The ethical decisions of computing professionals impact lives, businesses, and democracies—thus ethical codes like the BCS are vital safeguards.

IEEE Code of Ethics

What Is IEEE?

The **IEEE (Institute of Electrical and Electronics Engineers)** is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity. The **IEEE Code of Ethics** guides the professional behavior of its members.

IEEE's ethical code is a commitment to uphold the highest standards of integrity, responsible conduct, and public welfare.

Purpose of the IEEE Code

The Code of Ethics is designed to:

- Promote **honesty, integrity, and fairness**
- Protect the **public interest**
- Ensure **responsibility in engineering and technology practices**
- Guide professionals in making **ethical decisions**

IEEE Code of Ethics – Core Tenets

The IEEE Code of Ethics contains **ten fundamental principles** that members pledge to uphold:

IEEE Ethical Principles

1. **To hold paramount the safety, health, and welfare of the public**, and to disclose promptly factors that might endanger people or the environment.
2. **To avoid real or perceived conflicts of interest**, and to disclose them to affected parties when they exist.

3. **To be honest and realistic** in stating claims or estimates based on available data.
4. **To reject bribery**, and avoid improper influence in all its forms.
5. **To improve the understanding of technology**, its appropriate application, and its potential consequences.
6. **To maintain and improve our technical competence**, and to undertake technological tasks only if qualified by training or experience.
7. **To seek, accept, and offer honest criticism of technical work**, to acknowledge and correct errors, and to credit properly the contributions of others.
8. **To treat all persons fairly**, regardless of race, religion, gender, disability, age, or national origin.
9. **To avoid injuring others**, their property, reputation, or employment by false or malicious action.
10. **To assist colleagues and co-workers in their professional development**, and to support them in following this code of ethics.

Ethical Themes in IEEE Code

Theme	Description
Public Responsibility	Engineers must always consider public welfare, safety, and environmental sustainability.
Professional Integrity	Honesty in research, design, and reporting is non-negotiable.
Conflict of Interest	Must be transparently disclosed to maintain trust.
Inclusiveness	Ethical behavior includes fighting discrimination and promoting equity.

Lifelong Learning	Ethical professionals continue improving their competence.
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ACM Code of Ethics and Professional Conduct

What is ACM?

The **Association for Computing Machinery (ACM)** is the world's largest educational and scientific computing society. Its **Code of Ethics and Professional Conduct** establishes principles and responsibilities for all computing professionals.

The ACM Code promotes an ethical culture and provides a framework for responsible behavior in computing.

Purpose of the ACM Code

- Promote **honest, respectful, and responsible conduct**
- Protect **users, society, and stakeholders**
- Guide professionals in **ethical decision-making**
- Encourage **self-regulation** and professional integrity

Structure of the ACM Code

The code is structured around **four key sections**:

1. General Ethical Principles

These apply to all professional computing activities.

- 1.1 Contribute to society and human well-being

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- 1.2 Avoid harm
 - 1.3 Be honest and trustworthy
 - 1.4 Be fair and take action not to discriminate
 - 1.5 Respect the work required to produce new ideas
 - 1.6 Respect privacy
 - 1.7 Honor confidentiality

2. Professional Responsibilities

These address the conduct expected from professionals.

- 2.1 Strive to achieve high quality in professional work
- 2.2 Maintain high standards of professional competence and integrity
- 2.3 Know and respect existing laws
- 2.4 Accept and provide appropriate professional review
- 2.5 Give comprehensive and thorough evaluations
- 2.6 Honor contracts, agreements, and assigned responsibilities
- 2.7 Improve public understanding of computing
- 2.8 Access computing and communication resources only when authorized

3. Professional Leadership Principles

For those in leadership and management roles:

- 3.1 Ensure ethical design and implementation of systems
- 3.2 Manage personnel fairly and respectfully

- 3.3 Recognize and take responsibility for computing systems' societal impacts
- 3.4 Articulate and support ethical standards
- 3.5 Create opportunities for professional growth
- 3.6 Use care when modifying or retiring systems

4. Compliance with the Code

- Professionals must **uphold the code**
- **Violation** can lead to review or expulsion from ACM
- Encourages **reporting unethical conduct**

Scenario	Ethical Principle
Biased algorithm	1.2 Avoid harm, 1.4 Fairness
Data breach	1.6 Privacy, 2.3 Legal compliance
Misleading AI performance claims	1.3 Honesty, 2.1 Quality

Importance

Theme	Relevance
Public interest	Protects safety, well-being, and rights of users
Professional behavior	Promotes honesty, quality, and compliance



Social impact	Encourages foresight in design and deployment
Inclusive practices	Promotes fairness and non-discrimination
Security and privacy	Emphasizes data protection and confidentiality

ACM/IEEE Software Engineering Code of Ethics and Professional Practice

Purpose of the SE Code

The **Software Engineering Code of Ethics and Professional Practice** provides a **framework of ethical principles** and **professional standards** to guide software engineers in their work. It aims to:

- Ensure **public interest and safety**
- Promote **integrity and professionalism**
- Provide **standards for ethical judgment**
- Support **quality software development**

It applies to software engineers, developers, testers, project managers, and educators.

Eight Key Principles of the SE Code

The code is structured around **eight principles**, each representing a core ethical responsibility.

1. Public

Act consistently with the public interest.

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- Ensure software meets social needs and avoids harm
 - Respect privacy, safety, and welfare of users
 - Report potential risks and vulnerabilities honestly

2. Client and Employer

Act in the best interests of client and employer, consistent with the public interest.

- Maintain confidentiality
- Avoid deceptive claims
- Address software limitations openly
- Honor contracts and agreements

3. Product

Ensure that products meet the highest professional standards possible.

- Apply best practices in development and testing
- Identify, correct, and report defects
- Use appropriate tools and standards
- Document clearly and completely

4. Judgment

Maintain integrity and independence in professional judgment.

- Avoid conflicts of interest
- Base decisions on facts and well-established principles
- Do not let personal interest compromise quality or fairness



5. Management

Promote ethical approach to software development and maintenance.

- Provide leadership that encourages ethical conduct
- Ensure realistic project schedules and resources
- Allocate sufficient time for testing, documentation, and review

6. Profession

Advance the integrity and reputation of the profession.

- Support ethical learning and development
- Take responsibility for professional mistakes
- Share knowledge and improve the profession

7. Colleagues

Be fair and supportive to colleagues.

- Encourage ethical behavior among peers
- Give credit to others' work
- Avoid taking advantage of others
- Assist colleagues in professional growth

8. Self

Participate in lifelong learning and promote an ethical approach to practice.

- Keep skills up-to-date
- Stay informed of legal, technical, and ethical standards

- Act with dignity, honesty, and respect for the profession

Why does it matter?

Area	Contribution of SE Code
Public Protection	Encourages safety and risk mitigation
Quality Assurance	Emphasizes standards, testing, and documentation
Professional Integrity	Upholds fairness, honesty, and responsibility
Workplace Ethics	Guides collaboration, respect, and conflict resolution
Career Development	Promotes lifelong learning and ethical improvement

Example Application

Scenario: A software engineer discovers a critical security flaw in a healthcare application.

Ethical Response (based on SE Code):

- Report the issue (Principle 1: Public)
- Notify the employer and suggest a patch (Principle 2: Client and Employer)
- Document and verify the fix (Principle 3: Product)
- Do not hide the flaw due to pressure (Principle 4: Judgment)



Lecture 12: Accountability, Auditing and Social Implications of IT

Topics Covered: Accountability and Auditing, Social Application of Ethics.

Accountability and Auditing in Computing

Definition

Accountability

Accountability refers to the obligation of individuals, organizations, or systems to:

- **Accept responsibility** for their actions, and
- Be **answerable** for the outcomes—especially in case of failures, abuses, or policy violations.

Auditing

Auditing is the **systematic examination and verification** of records, logs, activities, and systems to:

- Ensure **compliance** with laws, regulations, and internal policies, and
- Detect **errors**, fraud, or unethical practices.

Importance in Computing and IT

- Detect **unauthorized access or misuse** of systems
- Monitor **system integrity** and performance
- Enforce **data protection regulations** (e.g., GDPR, HIPAA)
- Support **legal investigations** and accountability frameworks

- Build **trust** in digital platforms and IT infrastructure

Key Principles of Accountability

Principle	Description
Transparency	Actions and processes must be open and verifiable
Responsibility	Individuals must be held responsible for decisions made
Answerability	Must explain and justify actions taken, especially when systems fail
Liability	Legal or financial consequences must follow wrongful or negligent actions
Documentation	All important decisions and processes should be logged and archived

Types of Auditing

Type	Focus Areas
Security Audit	Access control, firewalls, intrusion detection systems
Compliance Audit	Adherence to laws (e.g., GDPR, SOX, HIPAA)
Software Audit	Licensing compliance, use of open-source components
Data Audit	Data accuracy, lifecycle, and protection
Process Audit	Software development life cycle (SDLC), project management

Tools and Techniques

- **Log analysis tools** (e.g., Splunk, Graylog)

- **Security Information and Event Management (SIEM)** systems
- **Audit trails** for databases and applications
- **Penetration testing** reports
- **Version control** and change logs (e.g., Git history)

Ethical and Legal Considerations

Issue	Concern
Privacy vs. Accountability	Excessive monitoring may violate user rights
Whistleblower protection	Auditors must be protected when reporting unethical behavior
Consent and transparency	Users must be informed about logging and monitoring practices
Bias in auditing tools	AI-based audits must be free of discrimination or flawed logic

Accountability in Software and Systems

- **Software developers** must take responsibility for bugs and design flaws
- **System administrators** are accountable for configurations and access management
- **Organizations** are legally liable for data breaches and non-compliance
- **Automated decision-making systems** must provide auditability for fairness and explainability

Example Scenarios

Scenario	Accountability/Auditing Role
Data breach in a hospital	Identify who accessed patient data and how

Bias in automated loan approval	Audit the algorithm and dataset
Failed IT project costing millions	Trace decision logs, approvals, and testing procedures
Employee abuse of privileged system access	Review access logs and session records

Social Application of Ethics in Computing

Introduction

Social application of ethics refers to the use of ethical principles to guide decision-making, behavior, and policy **in societal contexts**, especially when **technology** intersects with **human values, rights, and social systems**.

It is not only about what is legal, but what is **just, fair, and responsible** in the use of technology and information.

Why Social Ethics Matter in Computing

Modern technologies shape:

- **How we work**
- **How we communicate**
- **How decisions are made**
- **How we interact with government, education, healthcare, finance, and society**

As such, computing professionals are not only **technical workers** but also **social actors** who impact society at large.

Key Ethical Principles Applied Socially

Principle	Description
Justice & Fairness	Ensure equitable access, treatment, and distribution of technological benefits
Autonomy	Respect users' rights to make informed decisions (e.g., consent in data collection)
Non-maleficence	Avoid causing harm (e.g., AI bias, surveillance overreach)
Beneficence	Maximize benefits to society (e.g., e-learning, assistive tech)
Accountability	Individuals and institutions must be answerable for unethical practices

Key Areas of Social Application

Privacy and Surveillance

- **Ethical Dilemma:** Balancing national security with individual privacy
- **Example:** Use of CCTV, facial recognition, and online tracking

Digital Divide

- **Issue:** Unequal access to information and communication technologies (ICT)
- **Social Impact:** Widening educational and economic inequality

Freedom of Expression vs Harmful Content

- **Challenge:** Social media allows free speech but also spreads hate, fake news, and extremism

- **Ethical Balance:** Protect rights while minimizing societal harm

AI and Algorithmic Bias

- **Issue:** AI systems may unintentionally discriminate (e.g., hiring, policing, credit scoring)
- **Ethical Imperative:** Ensure fairness, transparency, and explainability in algorithmic decisions

Cyberbullying and Online Harassment

- **Social Harm:** Affects mental health, especially among youth
- **Ethical Duty:** Tech platforms must create safe online spaces and enforce policies

Social Responsibility of Tech Companies

- **Case Study:** Facebook–Cambridge Analytica scandal (data misuse for political manipulation)
- **Ethical Focus:** Transparency, informed consent, ethical product design

Case	Ethical Concern
Uber’s use of Greyball software	Deceived law enforcement; violated public trust
TikTok data handling controversy	Raised concerns over youth privacy and data sovereignty
AI used in judicial sentencing	Algorithmic bias influencing legal outcomes
Online learning tools	Beneficial socially, but risks surveillance and misuse

Emerging Social-Ethical Concerns

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- **Automation and Job Displacement** – Ethical planning needed for AI-driven unemployment
 - **Digital Identity & Reputation** – How online profiles impact real-life opportunities
 - **Virtual Reality & Metaverse** – New questions about behavior, consent, and identity
 - **Climate Impact of Technology** – Ethical design of sustainable digital infrastructure